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Percutaneous implantation of an annuloplasty structure

Abstract

During a percutaneous transcatheter procedure, an annuloplasty structure is secured at least partially around an annulus of a mitral valve of a subject by coupling a plurality of tissue anchors to the annulus such that (i) a longitudinal contracting member of the annuloplasty structure extends at least partially around the annulus, and (ii) the plurality of tissue anchors is distributed at least partially around the annulus. A longitudinal density of the anchors along a middle scallop of a posterior leaflet of the mitral valve is at least twice as great as a longitudinal density of the anchors along a lateral scallop of the posterior leaflet, and/or at least twice as great as a longitudinal density of the anchors along a medial scallop of the posterior leaflet. Thereafter, at least part of the annulus is contracted by contracting a longitudinal portion of the annuloplasty structure by tightening the longitudinal contracting member.

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8382829	12/2012	Call et al.	N/A	N/A
8388680	12/2012	Starksen et al.	N/A	N/A
8393517	12/2012	Milo	N/A	N/A
8419825	12/2012	Burgler et al.	N/A	N/A
8430926	12/2012	Kirson	N/A	N/A
8449573	12/2012	Chu	N/A	N/A
8449599	12/2012	Chau et al.	N/A	N/A
8454686	12/2012	Alkhatib	N/A	N/A
8460370	12/2012	Zakay	N/A	N/A
8460371	12/2012	Hlavka et al.	N/A	N/A
8475491	12/2012	Milo	N/A	N/A
8475525	12/2012	Maisano et al.	N/A	N/A
8480732	12/2012	Subramanian	N/A	N/A
8518107	12/2012	Tsukashima et al.	N/A	N/A
8523940	12/2012	Richardson et al.	N/A	N/A
8545553	12/2012	Zipory et al.	N/A	N/A
8551161	12/2012	Dolan	N/A	N/A
8585755	12/2012	Chau et al.	N/A	N/A
8591576	12/2012	Hasenkam et al.	N/A	N/A
8608797	12/2012	Gross et al.	N/A	N/A
8628569	12/2013	Benichou et al.	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8641727	12/2013	Starksen et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8652203	12/2013	Quadri et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685086	12/2013	Navia et al.	N/A	N/A
8728097	12/2013	Sugimoto et al.	N/A	N/A
8728155	12/2013	Montorfano et al.	N/A	N/A
8734467	12/2013	Miller et al.	N/A	N/A
8734699	12/2013	Heideman et al.	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8747463	12/2013	Fogarty et al.	N/A	N/A
8778021	12/2013	Cartledge	N/A	N/A
8784481	12/2013	Alkhatib et al.	N/A	N/A
8790367	12/2013	Nguyen et al.	N/A	N/A

8790394	12/2013	Miller et al.	N/A	N/A
8795298	12/2013	Hernlund et al.	N/A	N/A
8795355	12/2013	Alkhatib	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8795357	12/2013	Yohanan et al.	N/A	N/A
8808366	12/2013	Braido et al.	N/A	N/A
8808368	12/2013	Maisano et al.	N/A	N/A
8845717	12/2013	Khairkhahan et al.	N/A	N/A
8845723	12/2013	Spence et al.	N/A	N/A
8852261	12/2013	White	N/A	N/A
8852272	12/2013	Gross et al.	N/A	N/A
8858623	12/2013	Miller et al.	N/A	N/A
8864822	12/2013	Spence et al.	N/A	N/A
8870948	12/2013	Erzberger et al.	N/A	N/A
8870949	12/2013	Rowe	N/A	N/A
8888843	12/2013	Khairkhahan et al.	N/A	N/A
8889861	12/2013	Skead et al.	N/A	N/A
8894702	12/2013	Quadri et al.	N/A	N/A
8911461	12/2013	Traynor et al.	N/A	N/A
8911494	12/2013	Hammer et al.	N/A	N/A
8926696	12/2014	Cabiri et al.	N/A	N/A
8926697	12/2014	Gross et al.	N/A	N/A
8932343	12/2014	Alkhatib et al.	N/A	N/A
8932348	12/2014	Solem et al.	N/A	N/A
8940044	12/2014	Hammer et al.	N/A	N/A
8945211	12/2014	Sugimoto	N/A	N/A
8951285	12/2014	Sugimoto et al.	N/A	N/A
8951286	12/2014	Sugimoto et al.	N/A	N/A
8961595	12/2014	Alkhatib	N/A	N/A
8961602	12/2014	Kovach et al.	N/A	N/A
8979922	12/2014	Jayasinghe et al.	N/A	N/A
8992604	12/2014	Gross et al.	N/A	N/A
9005273	12/2014	Salahieh et al.	N/A	N/A
9011520	12/2014	Miller et al.	N/A	N/A
9011530	12/2014	Reich et al.	N/A	N/A
9023100	12/2014	Quadri et al.	N/A	N/A
9072603	12/2014	Tuval et al.	N/A	N/A
9107749	12/2014	Bobo et al.	N/A	N/A
9119719	12/2014	Zipory et al.	N/A	N/A
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9125742	12/2014	Yoganathan et al.	N/A	N/A
9138316	12/2014	Bielefeld	N/A	N/A
9173646	12/2014	Fabro	N/A	N/A
9180005	12/2014	Lashinski et al.	N/A	N/A
9180007	12/2014	Reich et al.	N/A	N/A
9192472	12/2014	Gross et al.	N/A	N/A
9198756	12/2014	Aklog et al.	N/A	N/A
9226825	12/2015	Starksen et al.	N/A	N/A

9265608	12/2015	Miller et al.	N/A	N/A
9326857	12/2015	Cartledge et al.	N/A	N/A
9414921	12/2015	Miller et al.	N/A	N/A
9427316	12/2015	Schweich, Jr. et al.	N/A	N/A
9474606	12/2015	Zipory et al.	N/A	N/A
9526613	12/2015	Gross et al.	N/A	N/A
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9579090	12/2016	Simms et al.	N/A	N/A
9693865	12/2016	Gilmore et al.	N/A	N/A
9724084	12/2016	Groothuis et al.	N/A	N/A
9730793	12/2016	Reich et al.	N/A	N/A
9788941	12/2016	Hacohen	N/A	N/A
9801720	12/2016	Gilmore et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of U.S. application Ser. No. 15/919,452, filed Mar. 13, 2018, which is: a divisional of U.S. application Ser. No. 14/589,100, filed Jan. 5, 2015 (now U.S. Pat. No. 9,918,840); and a continuation-in-part of U.S. application Ser. No. 15/474,543, filed Mar. 30, 2017 (now U.S. Pat. No. 10,470,882), which is a continuation of U.S. application Ser. No. 14/128,756, filed Feb. 6, 2014 (now U.S. Pat. No. 9,662,209), which is the US national phase of International Application

PCT/IL2012/000250, filed Jun. 21, 2012, which (a) is a continuation-in-part of U.S. application Ser. No. 13/167,444, filed Jun. 23, 2011 (now U.S. Pat. No. 9,011,530), (b) is a continuation-in-part of U.S. application Ser. No. 13/167,476, filed Jun. 23, 2011 (now U.S. Pat. No. 8,940,044), and (c) is a continuation-in-part of U.S. application Ser. No. 13/167,492, filed Jun. 23, 2011 (now U.S. Pat. No. 8,926,697), which is assigned to the assignee of the present application and is incorporated herein by reference.

FIELD OF THE APPLICATION

(1) Some applications of the present invention relate in general to valve repair, and more specifically to repair of an atrioventricular valve of a patient.

BACKGROUND OF THE APPLICATION

(2) Dilation of the annulus of the mitral valve prevents the valve leaflets from fully coapting when the valve is closed. Mitral regurgitation of blood from the left ventricle into the left atrium results in increased total stroke volume and decreased cardiac output, and ultimate weakening of the left ventricle secondary to a volume overload and a pressure overload of the left atrium. Dilation of the annulus is sometimes treated by annuloplasty, in which a partial or full ring is implanted around the annulus to cause the leaflets to coapt when the valve is closed.

SUMMARY

(3) In some applications of the present invention, an implantable structure is provided that comprises a flexible sleeve having first and second sleeve ends, a contracting assembly, and a plurality of tissue anchors. The contracting assembly is configured to longitudinally contract the sleeve, and comprises a contracting mechanism and a longitudinal contracting member having first and second member ends. The contracting mechanism is disposed longitudinally at a first site of the sleeve, and the second member end is coupled to the sleeve longitudinally at a second site longitudinally between the first site and the second sleeve end, exclusive. The contracting member also has a first member end portion, which extends from the first member end toward the second member end along only a longitudinal portion of the contracting member, and is coupled to the contracting mechanism. A first portion of the sleeve longitudinally extends from the first sleeve end toward the first site, and a second portion of the sleeve longitudinally extends from the second sleeve end toward the second site. The sleeve is arranged in a closed loop, such that the first and second portions of the sleeve together define a longitudinally overlapping portion of the sleeve. The implantable structure is configured such that the contracting assembly applies a longitudinal contracting force only between the first and the second sites, and not along the overlapping portion. The longitudinal contracting force longitudinally contracts at least a portion of the sleeve only between the first and the second sites, and not along the overlapping portion. Typically, the contracting member extends along neither the first nor the second portion of the sleeve.

(4) In some applications of the present invention, the contracting assembly includes one or more longitudinal contracting members coupled to the contracting mechanism. The implantable structure is placed completely around an annulus of an atrioventricular valve of a subject, such that none of the one or more longitudinal contracting members is positioned along an anterior portion of the annulus between fibrous trigones of the valve. The implantable structure is fastened to the annulus. The contracting assembly is then actuated to contract a longitudinal portion of the sleeve not positioned along the anterior portion of the annulus. Tightening of the implantable structure therefore tightens at least a portion of the posterior portion of the annulus, while preserving the length of the anterior portion of the annulus. (The anterior portion of the annulus should generally not be contracted because its tissue is part of the skeleton of the heart.) However, the portion of the sleeve deployed along the anterior portion of the annulus prevents dilation of the anterior annulus, because the sleeve is anchored at both ends of the anterior annulus, and the sleeve typically comprises a longitudinally non-extensible material. This deployment configuration may help prevent long-term resizing of annulus, especially the anterior annulus, which sometimes occurs

after implantation of partial annuloplasty rings, such as C-bands.

(5) In some applications of the present invention, one or more of the tissue anchors are coupled to the sleeve at respective third sites longitudinally between the second site and the second sleeve end, exclusive. Typically, the implantable structure is configured such that the contracting assembly applies a longitudinal contracting force only between the first and the second sites. The longitudinal contracting force contracts at least a portion of the sleeve only between the first and the second sites. Providing the one or more anchors beyond the ends of the contracting member generally distributes force applied by contraction of the contracting assembly over the tissue interfaces of these anchors. In contrast, in some configurations of the implantable structure in which anchors are not provided beyond the ends of the contracting member, the force applied by the contracting assembly is applied predominantly to the single anchor nearest the first end of the contracting member, and the single anchor nearest the second end of the contracting member.

(6) For some applications, at least two of the tissue anchors are coupled to the sleeve at respective third sites longitudinally between the second member end and the second sleeve end, exclusive. For some applications, the second site is at least 5 mm from the second sleeve end, measured when the sleeve is in a straight, relaxed, non-contracted state, such as at least 9 mm, e.g., at least 18 mm. For some applications, the second site is at a longitudinal distance from the second sleeve end, which distance is no greater than 30% of a total length of the sleeve, the distance and length measured when the sleeve is in the straight, relaxed, non-contracted state. For some applications, at least three of the tissue anchors are coupled to the sleeve alongside the contracting member, longitudinally between the first and second sites, exclusive. Typically, the sleeve is substantially longitudinally non-extensible.

(7) For some applications, the sleeve has first and second sleeve ends, and first and second portions that longitudinally extend from the first and the second sleeve ends, respectively. The sleeve is arranged in a closed loop, such that the first and second portions of the sleeve together define a longitudinally overlapping portion of the sleeve positioned at least partially along the anterior portion of the annulus, and none of the one or more longitudinal contracting members is positioned along the overlapping portion of the sleeve. For some applications, at least one of the tissue anchors penetrates both the first and second portions of the sleeve at the overlapping portion. Such a mutual anchor helps ensure that the first and second portions remain tightly coupled together and to the tissue, so that the sleeve retains its closed loop shape. Alternatively, for some applications, the sleeve is shaped so as to define an integrally closed loop having no sleeve ends.

(8) The implantable structure, when in this closed-loop configuration, is deployed around the entire annulus of the native valve, including an anterior portion of the annulus (on the aortic side of the valve) between the fibrous trigones. Typically, the contracting member does not extend along the portion of the sleeve deployed along the anterior portion of the annulus, and thus does not extend along the first portion, the second portion, or the overlapping portion of the sleeve. The portion of the sleeve deployed along the anterior portion of the annulus (between the trigones) is thus non-contractible. As mentioned above, tightening of the implantable structure therefore tightens the posterior portion of the annulus, while preserving the length of the anterior portion of the annulus. For some applications, this deployment configuration may also help achieve a closed loop that serves as a base ring to which a prosthetic valve is coupled.

(9) In some applications of the present invention, the implantable structure further comprises an elongated linking member, which is positioned along an anterior portion of the annulus, so as to join the ends of the implantable structure in a complete loop. Over time after implantation, the linking member becomes fixed to the anterior portion of the annulus, thereby helping prevent long-term dilation of the anterior annulus. Typically, at least a portion of the linking member is disposed within and covered by the sleeve, into and/or over which fibrous tissue grows over time, helping anchor the linking member to tissue of the anterior annulus. Typically, in this configuration of the implantable structure, none of the anchors is coupled to the anterior portion of the annulus.

(10) A first end of the linking member is typically fixed between 2 and 6 cm from a first end of the sleeve. A second end of the linking member is positioned within 1.5 cm of the same end of the sleeve, either protruding from the end of the sleeve, or recessed within the sleeve. The second end of the linking member comprises (e.g., is shaped so as to define) a first coupling element. The implantable structure further comprises a second coupling element, which is configured to be coupleable to the first coupling element. The second coupling element is coupled to the implantable structure within 1.5 cm of the second end of the sleeve. The second coupling element may be coupled to the housing, directly to the sleeve, or otherwise coupled to the implantable structure. Typically, the linking member is substantially longitudinally non-extensible, i.e., its length is fixed.

(11) For some applications, the linking member is configured as a spring, which is typically curved, so as to be elastic in a radial direction, i.e., to be compressible like a bow or deflected beam. In these applications, the linking member is oriented such that it is pressed by elasticity against the anterior portion of the mitral annulus, i.e., the outer wall of the aorta, thereby holding the sleeve covering the linking member against the aortic wall. For some applications, at least two of the tissue anchors are coupled to the sleeve at respective, different longitudinal sites alongside the linking member, within 6 cm of the first end of the linking member. These tissue anchors may help set the proper direction of curvature of the linking member, for applications in which the linking member is curved.

(12) For some applications, the implantable structure further comprises an elongated radial-force application element, which is disposed entirely within a first longitudinal portion of the sleeve. The elongated radial-force application element is configured to apply a force against a wall of the first longitudinal portion of the sleeve in at least one radially-outward direction. The applied force pushes the first longitudinal portion of the sleeve against tissue of the left atrium, such as against tissue of the annulus and/or the atrial wall, so as to inhibit blood flow between the sleeve and the tissue. It is generally desirable to inhibit blood flow between the sleeve and the annulus on anterior side, to avoid creating turbulence. When implanting the implantable structure, the elongated radial-force application element is placed along the anterior portion of the annulus, between the fibrous trigones.

(13) For some applications, the elongated radial-force application element comprises a springy element. For some applications, at least a portion of the springy element is curved at least partially about an inner surface of the wall of the sleeve.

(14) For some applications, the elongated radial-force application element is rotationally asymmetric and not helically symmetric. For other applications, the elongated radial-force application element is helically symmetric; for these applications, the springy element typically comprises a coiled spring.

(15) For some applications, the sleeve has first and second sleeve ends. For some applications, the elongated radial-force application element has (a) a first radial-force-application-element longitudinal end that is between 2 and 6 cm from the first sleeve end, measured when the sleeve is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end that is within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended.

(16) For some applications, the annuloplasty ring further comprises (a) a first coupling element, which is coupled to the annuloplasty ring within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended, and (b) a second coupling element. The second coupling element is configured to be coupleable to the first coupling element, and is fixed to the implantable structure (e.g., the annuloplasty ring) within 1.5 cm of the second sleeve end, measured when the sleeve is fully longitudinally extended. For some applications, at least one of the first and second coupling elements comprises a hook.

(17) For some applications, the contracting mechanism (e.g., the housing thereof) is fixed along the sleeve within 30 mm, such as within 15 mm, of the second sleeve end (i.e., the same end of the

sleeve near which the second coupling element is coupled), measured when the sleeve is fully longitudinally extended. For example, the contracting mechanism (e.g., the housing thereof) may be fixed at the second sleeve end. Alternatively, for some applications, the contracting mechanism (e.g., the housing thereof) is fixed at least 5 mm from the second sleeve end, e.g., between 5 and 30 mm, such as between 5 and 15 mm, from the second sleeve end. The second coupling element may be coupled to the contracting mechanism (e.g., to the housing).

(18) For some applications, the annuloplasty ring further comprises a substantially longitudinally non-extensible linking member, i.e., a length thereof is substantially constant, i.e., cannot be longitudinally stretched, under normal usage conditions. The linking member typically helps prevent long-term dilation of the anterior annulus. The linking member is typically configured not to apply any force to the wall of the first longitudinal portion of the sleeve. Typically, the linking member is not configured as a spring.

(19) For some applications, at least the first longitudinal portion of the sleeve is substantially longitudinally non-extensible, i.e., a length thereof is substantially constant, i.e., cannot be longitudinally stretched, under normal usage conditions. In these applications, the first longitudinal portion typically helps prevent long-term dilation of the anterior annulus. For some applications, the first coupling element is fixed to the wall of the sleeve within 1.5 cm of first sleeve end **51**, measured when the sleeve is fully longitudinally extended. The implantable structure typically does not comprise the linking member in these applications. In these applications, at least the first longitudinal portion of the sleeve is substantially longitudinally non-extensible, and the first longitudinal portion typically helps prevent long-term dilation of the anterior annulus.

(20) For some applications, during placement, after fastening the sleeve to the portion of the annulus, the healthcare professional twists the first longitudinal portion of the sleeve. Optionally, such twisting may serve one or both of the following purposes: (1) the twisting may store energy in the springy element for exertion of torque against the wall of the sleeve, and (2) the twisting may rotationally align the springy element in the desired radial direction. Alternatively or additionally to twisting for the first of these purposes, the springy element may be pre-loaded (twisted) to store energy before implantation in the subject, such as immediately before implantation or during manufacture.

(21) For some applications, the sleeve is fastened to the annulus by coupling a plurality of tissue anchors to the annulus. The tissue anchors are coupled with: a first non-zero longitudinal density along a posterior portion of the annulus between the left and right fibrous trigones of the annulus, including the trigones, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the posterior portion of the annulus divided by (b) a length of the posterior portion of the annulus (measured along the annulus), and a second non-zero longitudinal density along the anterior portion of the annulus between the left and right fibrous trigones of the annulus, not including the trigones, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the anterior portion of the annulus divided by (b) a length of the anterior portion of the annulus (measured along the annulus).

(22) The first longitudinal density is greater than the second longitudinal density. For some applications, the first longitudinal density is at least twice the second longitudinal density, such as at least 2.5 the second longitudinal density, e.g., at least 3 times the second longitudinal density. After the tissue anchors are fastened to the annulus, a longitudinal portion of the sleeve is contracted, such as by causing the longitudinal contracting member to apply a force to the longitudinal portion of the sleeve, such as by actuating the contracting assembly.

(23) For some applications, the sleeve is fastened to the annulus by coupling a plurality of tissue anchors to the annulus, including first, second, and third tissue anchors, as follows: one or more first tissue anchors are coupled to the annulus along a lateral scallop (P1) of the posterior leaflet, with a first longitudinal density, which density is equal to (a) a number of the first tissue anchors coupled to the annulus along the lateral scallop (P1) divided by (b) a length of the lateral scallop

(P1) along the annulus, a plurality of second tissue anchors (e.g., at least 3 tissue anchors) are coupled to the annulus along a middle scallop (P2) of the posterior leaflet, with a second longitudinal density, which density is equal to (a) a number of the second tissue anchors coupled to the annulus along the middle scallop (P2) divided by (b) a length of the middle scallop (P2) along the annulus, and one or more third tissue anchors are coupled to the annulus along a medial scallop (P3) of the posterior leaflet, with a third longitudinal density, which density is equal to (a) a number of the third tissue anchors coupled to the annulus along the medial scallop (P3) divided by (b) a length of the medial scallop (P3) along the annulus.

(24) The longitudinal densities are characterized by at least one of the following: (a) the second longitudinal density is at least twice the first longitudinal density, and (b) the second longitudinal density is at least twice the third longitudinal density. For some applications, both (a) the second longitudinal density is at least twice the first longitudinal density, and (b) the second longitudinal density is at least twice the third longitudinal density.

(25) For some applications, the tissue anchors, including the second tissue anchors, comprise respective anchor heads and tissue coupling elements. Typically, the anchor heads are circular; alternatively, they have another shape, such as of an ellipse or a polygon (e.g., a hexagon or a square). The plurality of tissue anchors are coupled to the annulus such that, after the longitudinal portion of the sleeve has been contracted, each of the anchor heads of at least two of the second tissue anchors coupled along the middle scallop (P2) touches at least one longitudinally-adjacent anchor head; for example, each of the anchor heads of at least three of tissue anchors touches at least one longitudinally-adjacent anchor head **320**.

(26) Typically, before the longitudinal portion of the sleeve has been contracted, the anchor heads of the at least two of the second tissue anchors do not touch any longitudinally-adjacent the anchor heads. Before the longitudinal portion of the sleeve has been contracted, the anchors are coupled to the sleeve and tissue at distances between the anchors that are less than the planned distances that the anchors move toward each other during contraction of the longitudinal portion of the sleeve. As a result, the anchor heads touch each other upon such contraction.

(27) This touching of the longitudinally-adjacent anchors heads inhibits longitudinal contraction of the sleeve in the longitudinal area of these anchors, so as to facilitate reshaping of the annulus in a desired manner. These longitudinally-adjacent the anchor heads thus are dual-function, and serve to both anchor their respective anchors to the sleeve and to inhibit contraction of the sleeve.

(28) For some applications, the plurality of tissue anchors is coupled to the annulus such that, after the longitudinal portion of the sleeve has been contracted: none of the anchor heads of the first tissue anchors coupled along the lateral scallop (P1) touches any of the other anchor heads of the tissue anchors; and/or none of the anchor heads of the third tissue anchors coupled along the medial scallop (P3) touches any of the other anchor heads of the tissue anchors.

(29) For some applications, the plurality of tissue anchors are coupled to the annulus such that, after the longitudinal portion of the sleeve has been contracted: a first number of the anchor heads of the first tissue anchors coupled along the lateral scallop (P1) touch at least one longitudinally-adjacent anchor head, and (b) a second number of the anchors heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, the second number greater than the first number; and/or a second number of the anchor heads of the second tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, and (b) a third number of the anchors heads of the third tissue anchors coupled along the medial scallop (P3) touch at least one longitudinally-adjacent anchor head, the second number greater than the third number.

(30) For some applications, the sleeve is fastened to the annulus by coupling a plurality of tissue anchors to the annulus, such that: a first set of exactly three of the tissue anchors is disposed in succession along a first portion of the longitudinal contracting member with a first distance between longitudinal-end tissue anchors of the first set, measured along the annulus, and a second

set of exactly three of the tissue anchors is disposed in succession along a second portion of the longitudinal contracting member with a second distance between longitudinal-end tissue anchors of the second set, measured along the annulus,

(31) The first distance equals at least twice the second distance, such as at least 2.5 times the second distance, e.g., at least 3 times the second distance. The first distance is measured between closest portions of the longitudinal-end tissue anchors of the first set, and the second distance is measured between closest portions of the longitudinal-end tissue anchors of the second set. The first and second sets do not share any common tissue anchors. After the tissue anchors are fastened to the annulus, a longitudinal portion of the sleeve is contracted. Providing the greater number of anchoring points with the second set better distributes forces among the anchors of this set.

(32) For some applications, the contracting mechanism comprises a rotatable structure, and a housing in which the rotatable structure is positioned. The contracting mechanism and the longitudinal contracting member are arranged such that rotation of the rotatable structure contracts the implantable structure. Typically, an anchor deployment manipulator is advanced into a lumen of the sleeve, and, from within the lumen, deploys the anchors through a wall of the sleeve and into cardiac tissue, thereby anchoring the sleeve around a portion of a valve annulus.

(33) For some applications, the implantable structure comprises an adjustable annuloplasty ring for repairing a dilated valve annulus of an atrioventricular valve, such as a mitral valve. The annuloplasty ring may be used for treating functional mitral regurgitation (FMR) or degenerative mitral valve disease. For other applications, a prosthetic heart valve is further provided, which is configured to be coupled to the sleeve.

(34) For some applications in which the implantable structure is implanted around the annulus of a valve, the implantable structure may be advanced toward the annulus of a valve in any suitable procedure, e.g., a transcatheter procedure, a percutaneous procedure, a minimally invasive procedure, or an open heart procedure.

(35) There is therefore provided, in accordance with an application of the present invention, a method including: providing an annuloplasty ring, which includes (a) a flexible sleeve, and (b) a contracting assembly; during a percutaneous transcatheter procedure, placing the flexible sleeve entirely around an annulus of a mitral valve of a subject in a closed loop; fastening the sleeve to the annulus by coupling a plurality of tissue anchors to a posterior portion of the annulus, without coupling any tissue anchors to an anterior portion of the annulus between left and right fibrous trigones of the annulus; and thereafter, contracting a longitudinal portion of the sleeve.

(36) For some applications, the contracting assembly further includes a longitudinal contracting member and a locking mechanism, and the method further includes, after contracting the longitudinal portion of the sleeve, locking the longitudinal contracting member with respect to the contracting assembly using the locking mechanism.

(37) For some applications, contracting the longitudinal portion of the sleeve includes actuating the contracting assembly to contract the longitudinal portion of the sleeve.

(38) For some applications, providing the annuloplasty ring includes providing the annuloplasty ring in which the sleeve is shaped so as to define an integrally closed loop having no sleeve ends.

(39) For some applications, the sleeve has first and second sleeve ends, and placing the sleeve includes introducing the flexible sleeve into a left atrium while the first and the second sleeve ends are not coupled to each other; and thereafter, in the left atrium, arranging the flexible sleeve entirely around the annulus to form the closed loop.

(40) For some applications, the annuloplasty ring further includes an elongated linking member, which is coupled to and disposed within the sleeve, and placing the flexible sleeve entirely around the annulus includes placing the linking member along the anterior portion of the annulus.

(41) For some applications, the linking member is configured as a spring. For some applications, the linking member is curved. For some applications, the linking member has a length of between 2 and 6 cm. For some applications, the linking member includes metal. For some applications, the

linking member is substantially longitudinally non-extensible.

(42) For some applications: the linking member includes a first coupling element, the annuloplasty ring includes a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the annuloplasty ring within 1.5 cm of one of the first and the second sleeve ends, measured when the sleeve is fully longitudinally extended, the first and the second coupling elements are configured to provide an adjustable-length connection between the linking member and the one of the first and the second sleeve ends, and placing the linking member along the anterior portion of the annulus includes setting an effective length of the linking member while coupling the first and the second coupling elements together.

(43) For some applications: the linking member is disposed within a longitudinal portion of the sleeve, the annuloplasty ring further includes an elongated radial-force application element, which is disposed within the longitudinal portion of the sleeve, and placing the linking member includes placing the elongated radial-force application element along the anterior portion of the annulus, such that the elongated radial-force application element applies a force against a wall of the longitudinal portion of the sleeve in at least one radially-outward direction.

(44) For some applications, placing the elongated radial-force application element includes placing the elongated radial-force application element along the anterior portion of the annulus, such that the elongated radial-force application element pushes the longitudinal portion of the sleeve against atrial tissue.

(45) For some applications, the elongated radial-force application element is springy.

(46) For some applications, the elongated radial-force application element includes an inflatable element.

(47) For some applications, the linking member is not configured as a spring.

(48) For some applications, placing the linking member includes placing the linking member such that the linking member does not apply any force to the wall of the longitudinal portion of the sleeve.

(49) For some applications, at least 90% of a length of the linking member is straight when in a resting state.

(50) For some applications, the linking member is substantially longitudinally non-extensible.

(51) For some applications, the elongated radial-force application element has a length of between 2 and 6 cm, measured when the sleeve is fully longitudinally extended.

(52) For some applications: the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, the contracting assembly includes (a) a contracting mechanism, and (b) a longitudinal contracting member, which is arranged along a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve, and the elongated radial-force application element is disposed entirely within the first longitudinal portion of the sleeve.

(53) There is further provided, in accordance with an application of the present invention, a method including: providing an annuloplasty ring, which includes (a) a flexible sleeve, and (b) a contracting assembly; during a percutaneous transcatheter procedure, placing the flexible sleeve entirely around an annulus of a mitral valve of a subject in a closed loop; fastening the sleeve to the annulus by coupling a plurality of tissue anchors to the annulus, with: a first non-zero longitudinal density of the tissue anchors along a posterior portion of the annulus between left and right fibrous trigones of the annulus, including the trigones, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the posterior portion of the annulus divided by (b) a length of the posterior portion of the annulus, and a second non-zero longitudinal density of the tissue anchors along an anterior portion of the annulus between left and right fibrous trigones of the annulus, not including the trigones, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the anterior portion of the annulus divided by (b) a length of the anterior portion of the annulus, wherein the first longitudinal density is greater than the second

longitudinal density; and thereafter, contracting a longitudinal portion of the sleeve.

(54) For some applications, the contracting assembly further includes a longitudinal contracting member and a locking mechanism, and the method further includes, after contracting the longitudinal portion of the sleeve, locking the longitudinal contracting member with respect to the contracting assembly using the locking mechanism.

(55) For some applications, contracting the longitudinal portion of the sleeve includes actuating the contracting assembly to contract the longitudinal portion of the sleeve.

(56) For some applications, the first longitudinal density is at least twice the second longitudinal density.

(57) For some applications, providing the annuloplasty ring includes providing the annuloplasty ring in which the sleeve is shaped so as to define an integrally closed loop having no sleeve ends.

(58) For some applications, the sleeve has first and second sleeve ends, and placing the flexible sleeve includes introducing the flexible sleeve into a left atrium while the first and the second sleeve ends are not coupled to each other; and thereafter, in the left atrium, arranging the flexible sleeve entirely around the annulus to form the closed loop.

(59) For some applications, the annuloplasty ring further includes an elongated linking member, which is coupled to and disposed within the sleeve, and placing the flexible sleeve entirely around the annulus includes placing the linking member along the anterior portion of the annulus.

(60) For some applications, the linking member has a length of between 2 and 6 cm.

(61) For some applications, the linking member includes metal.

(62) For some applications, the linking member is substantially longitudinally non-extensible.

(63) For some applications: the linking member includes a first coupling element, the annuloplasty ring includes a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the annuloplasty ring within 1.5 cm of one of the first and the second sleeve ends, measured when the sleeve is fully longitudinally extended, the first and the second coupling elements are configured to provide an adjustable-length connection between the linking member and the one of the first and the second sleeve ends, and placing the linking member along the anterior portion of the annulus includes setting an effective length of the linking member while coupling the first and the second coupling elements together.

(64) For some applications: the linking member is disposed within a longitudinal portion of the sleeve, the annuloplasty ring further includes an elongated radial-force application element, which is disposed within the longitudinal portion of the sleeve, and placing the linking member includes placing the elongated radial-force application element along the anterior portion of the annulus, such that the elongated radial-force application element applies a force against a wall of the longitudinal portion of the sleeve in at least one radially-outward direction.

(65) For some applications, placing the elongated radial-force application element includes placing the elongated radial-force application element along the anterior portion of the annulus, such that the elongated radial-force application element pushes the longitudinal portion of the sleeve against atrial tissue.

(66) For some applications, the elongated radial-force application element is springy.

(67) For some applications, the elongated radial-force application element includes an inflatable element.

(68) For some applications, the linking member is not configured as a spring.

(69) For some applications, placing the linking member includes placing the linking member such that the linking member does not apply any force to the wall of the longitudinal portion of the sleeve.

(70) For some applications, at least 90% of a length of the linking member is straight when in a resting state.

(71) For some applications, the linking member is substantially longitudinally non-extensible.

(72) For some applications, the elongated radial-force application element has a length of between

2 and 6 cm, measured when the sleeve is fully longitudinally extended.

(73) For some applications: the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, the contracting assembly includes (a) a contracting mechanism, and (b) a longitudinal contracting member, which is arranged along a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve, and the elongated radial-force application element is disposed entirely within the first longitudinal portion of the sleeve.

(74) There is still further provided, in accordance with an application of the present invention, a method including: providing an annuloplasty ring, which includes (a) a flexible sleeve, and (b) a contracting assembly; during a percutaneous transcatheter procedure, placing the flexible sleeve at least partially around an annulus of a mitral valve of a subject; fastening the sleeve to the annulus by coupling a plurality of tissue anchors to the annulus, with: a first longitudinal density of the tissue anchors along a lateral scallop (P1) of a posterior leaflet of the mitral valve, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the lateral scallop (P1) divided by (b) a length of the lateral scallop (P1) along the annulus, a second longitudinal density of the tissue anchors along a middle scallop (P2) of the posterior leaflet, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the middle scallop (P2) divided by (b) a length of the middle scallop (P2) along the annulus, and a third longitudinal density of the tissue anchors along a medial scallop (P3) of the posterior leaflet, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the medial scallop (P3) divided by (b) a length of the medial scallop (P3) along the annulus, wherein the longitudinal densities are characterized by at least one of the following: (a) the second longitudinal density is at least twice the first longitudinal density, and (b) the second longitudinal density is at least twice the third longitudinal density; and thereafter, contracting a longitudinal portion of the sleeve.

(75) For some applications, the contracting assembly further includes a longitudinal contracting member and a locking mechanism, and the method further includes, after contracting the longitudinal portion of the sleeve, locking the longitudinal contracting member with respect to the contracting assembly using the locking mechanism.

(76) For some applications, contracting the longitudinal portion of the sleeve includes actuating the contracting assembly to contract the longitudinal portion of the sleeve.

(77) For some applications, both (a) the second longitudinal density is at least twice the first longitudinal density, and (b) the second longitudinal density is at least twice the third longitudinal density.

(78) For some applications, the second longitudinal density is at least twice the first longitudinal density.

(79) For some applications, the second longitudinal density is at least twice the third longitudinal density.

(80) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling at least 3 tissue anchors to the annulus along the middle scallop (P2).

(81) For some applications, the tissue anchors have respective anchor heads, and coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve, each of the anchor heads of at least two of the tissue anchors coupled along the middle scallop (P2) touches at least one longitudinally-adjacent anchor head.

(82) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, before contracting the longitudinal portion of the sleeve, the anchor heads of the at least two of the tissue anchors do not touch the at least one longitudinally-adjacent anchor head.

(83) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal

portion of the sleeve, each of the anchor heads of at least three of the tissue anchors coupled along the middle scallop (P2) touches at least one longitudinally-adjacent anchor head.

(84) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve, none of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touches any of the other anchor heads of the tissue anchors.

(85) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve, none of the anchor heads of the tissue anchors coupled along the medial scallop (P3) touches any of the other anchor heads of the tissue anchors.

(86) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve, (a) none of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touches any of the other anchor heads of the tissue anchors, and (b) none of the anchor heads of the tissue anchors coupled along the medial scallop (P3) touches any of the other anchor heads of the tissue anchors.

(87) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve, (a) a first number of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touch at least one longitudinally-adjacent anchor head, and (b) a second number of the anchor heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, the second number greater than the first number.

(88) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve, (a) a second number of the anchor heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, and (b) a third number of the anchor heads of the tissue anchors coupled along the medial scallop (P3) touch at least one longitudinally-adjacent anchor head, the second number greater than the third number.

(89) For some applications, coupling the plurality of tissue anchors to the annulus includes coupling the plurality of tissue anchors to the annulus such that, after contracting the longitudinal portion of the sleeve: a first number of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touch at least one longitudinally-adjacent anchor head, a second number of the anchor heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, and a third number of the anchor heads of the tissue anchors coupled along the medial scallop (P3) touch at least one longitudinally-adjacent anchor head, the second number greater than the first number, and the second number greater than the third number.

(90) For some applications, the sleeve has first and second sleeve ends, and placing the sleeve includes introducing the flexible sleeve into a left atrium while the first and the second sleeve ends are not coupled to each other.

(91) For some applications, placing the sleeve includes arranging the sleeve entirely around the annulus to form a closed loop, after introducing the flexible sleeve into the left atrium while the first and the second sleeve ends are not coupled to each other.

(92) For some applications, providing the annuloplasty ring includes providing the annuloplasty ring in which the sleeve is shaped so as to define an integrally closed loop having no sleeve ends.

(93) There is additionally provided, in accordance with an application of the present invention, a method including: providing an annuloplasty ring, which includes (a) a flexible sleeve and (b) a contracting assembly, which includes a longitudinal contracting member; during a percutaneous transcatheter procedure, placing the flexible sleeve at least partially around an annulus of a mitral valve of a subject; fastening the sleeve to the annulus by coupling a plurality of tissue anchors to the annulus, such that: a first set of exactly three of the tissue anchors is disposed in succession

along the longitudinal contracting member with a first distance between longitudinal-end tissue anchors of the first set, measured along the annulus, and a second set of exactly three of the tissue anchors is disposed in succession along the longitudinal contracting member with a second distance between longitudinal-end tissue anchors of the second set, measured along the annulus, wherein the first distance equals at least twice the second distance, and wherein the first and the second sets do not share any common tissue anchors; and thereafter, contracting a longitudinal portion of the sleeve by causing the longitudinal contracting member to apply a contracting force to the longitudinal portion of the sleeve.

(94) For some applications, the contracting assembly further includes a locking mechanism, and the method further includes, after contracting the longitudinal portion of the sleeve, locking the longitudinal contracting member with respect to the contracting assembly using the locking mechanism.

(95) For some applications, contracting the longitudinal portion of the sleeve includes actuating the contracting assembly to contract the longitudinal portion of the sleeve by causing the longitudinal contracting member to apply the contracting force to the longitudinal portion of the sleeve.

(96) There is yet additionally provided, in accordance with an application of the present invention, apparatus including an annuloplasty ring, which includes: a flexible sleeve, having first and second sleeve ends; a contracting assembly; a first coupling element, which is coupled to the annuloplasty ring within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended; a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the annuloplasty ring within 1.5 cm of the second sleeve end, measured when the sleeve is fully longitudinally extended; and an elongated springy element, which is disposed entirely within a longitudinal portion of the sleeve, wherein the springy element has (a) a first springy-element longitudinal end that is between 2 and 6 cm from the first sleeve end, measured when the sleeve is fully longitudinally extended, and (b) a second springy-element longitudinal end that is within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended, wherein the springy element is configured to press the longitudinal portion of the sleeve against tissue.

(97) For some applications, the contracting assembly further includes a longitudinal contracting member and a locking mechanism, which is configured to lock the longitudinal contracting member with respect to the contracting assembly.

(98) For some applications, the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, and the contracting assembly is configured to contract at least a portion of a second longitudinal portion of the sleeve, which second longitudinal portion is entirely longitudinally distinct from the first longitudinal portion.

(99) For some applications, a first end of the elongated springy element includes the first coupling element.

(100) There is also provided, in accordance with an application of the present invention, a method including: providing an annuloplasty ring, which includes (a) a flexible sleeve, having first and second sleeve ends, (b) a contracting assembly, (c) a first coupling element, which is coupled to the annuloplasty ring within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended, (d) a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the annuloplasty ring within 1.5 cm of the second sleeve end, measured when the sleeve is fully longitudinally extended, and (e) an elongated springy element, which is disposed entirely within a first longitudinal portion of the sleeve, wherein the springy element has (a) a first springy-element longitudinal end that is between 2 and 6 cm from the first sleeve end, measured when the sleeve is fully longitudinally extended, and (b) a second springy-element longitudinal end that is within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended; during a percutaneous transcatheter procedure, placing the flexible sleeve around a portion of an annulus of an atrioventricular valve of a subject, which

portion includes a posterior portion of the annulus; placing the first longitudinal portion of the sleeve along an anterior portion of the annulus between fibrous trigones of the valve; fastening the flexible sleeve to the portion of the annulus, such that the springy element presses the first longitudinal portion of the sleeve against tissue; coupling the first and the second coupling elements together; and contracting at least a portion of a second longitudinal portion of the sleeve, which second longitudinal portion is entirely longitudinally distinct from the first longitudinal portion.

(101) For some applications, the contracting assembly further includes a locking mechanism, and the method further includes, after contracting the at least a portion of the second longitudinal portion of the sleeve, locking the longitudinal contracting member with respect to the contracting assembly using the locking mechanism.

(102) For some applications, contracting the at least a portion of the second longitudinal portion of the sleeve includes actuating the contracting assembly to contract the at least a portion of the second longitudinal portion of the sleeve.

(103) For some applications, a first end of the elongated springy element includes the first coupling element.

(104) There is further provided, in accordance with an application of the present invention, apparatus including an annuloplasty ring, which includes: a flexible sleeve; and an elongated radial-force application element, which (a) is disposed entirely within a longitudinal portion of the sleeve, (b) which has a length of no more than 6 cm, measured when the sleeve is fully longitudinally extended, and (c) is configured to apply a force against a wall of the longitudinal portion of the sleeve in at least one radially-outward direction.

(105) For some applications, the elongated radial-force application element is rotationally asymmetric and not helically symmetric.

(106) For some applications, the elongated radial-force application element is configured to apply the force against the wall around less than 100% of a perimeter of the wall.

(107) For some applications, the elongated radial-force application element is configured to apply the force against the wall around less than 50% of the perimeter of the wall.

(108) For some applications, the elongated radial-force application element is configured to apply the force with a variation of less than 20% along a length of the elongated radial-force application element.

(109) For some applications, the sleeve has first and second sleeve ends.

(110) For some applications, the annuloplasty ring further includes: a first coupling element, which is coupled to the annuloplasty ring within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended; and a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the annuloplasty ring within 1.5 cm of the second sleeve end, measured when the sleeve is fully longitudinally extended.

(111) For some applications, the elongated radial-force application element has (a) a first radial-force-application-element longitudinal end that is between 2 and 6 cm from the first sleeve end, measured when the sleeve is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end that is within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended.

(112) For some applications, the sleeve is shaped so as to define an integrally closed loop having no sleeve ends.

(113) For some applications: the annuloplasty ring further includes a contracting assembly, which includes a housing that is fixed to the sleeve, and the elongated radial-force application element has (a) a first radial-force-application-element longitudinal end that is between 2 and 6 cm from the housing, measured when the sleeve is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end that is within 1.5 cm of the housing, measured when the sleeve is fully longitudinally extended.

- (114) For some applications, the elongated radial-force application element is configured to push the longitudinal portion of the sleeve against atrial tissue.
- (115) For some applications, the annuloplasty ring further includes a substantially longitudinally non-extensible linking member, which has first and second linking-member ends and is at least partially disposed within the longitudinal portion of the sleeve, and the second linking-member end includes the first coupling element.
- (116) For some applications, the linking member has a length of between 2 and 6 cm.
- (117) For some applications, at least the longitudinal portion of the sleeve is substantially longitudinally non-extensible, and the first coupling element is fixed to the wall of the sleeve within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended.
- (118) For some applications, the elongated radial-force application element includes a springy element.
- (119) For some applications, where at least a portion of the springy element is curved at least partially about an inner surface of the wall of the sleeve.
- (120) For some applications, at least a portion of the springy element is serpentine.
- (121) For some applications, the at least a portion of the springy element is curved at least partially about the inner surface of the wall in a single circumferential direction.
- (122) For some applications, at least a first portion of the springy element is curved at least partially about the inner surface of the wall in a first circumferential direction, and at least a second portion of the springy element is curved at least partially about the inner surface of the wall in a second circumferential direction circumferentially opposite the first circumferential direction.
- (123) For some applications, at least a portion of the springy element is serpentine.
- (124) For some applications, springy element includes a coiled spring.
- (125) For some applications, the elongated radial-force application element includes an inflatable element.
- (126) For some applications, the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, and the annuloplasty ring further includes a longitudinal contracting member, which is arranged only along a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve.
- (127) For some applications, the annuloplasty ring further includes a contracting assembly, which includes the longitudinal contracting member and a contracting mechanism.
- (128) For some applications, a first average internal diameter of the first longitudinal portion of the sleeve is greater than a second average internal diameter of the second longitudinal portion of the sleeve, when both the first and the second longitudinal portions are fully radially expanded.
- (129) For some applications, the first longitudinal portion of the sleeve is radially elastic, and the second longitudinal portion of the sleeve is substantially radially non-extensible.
- (130) For some applications, the first and the second longitudinal portions of the sleeve are substantially longitudinally non-extensible.
- (131) For some applications, the first and the second longitudinal portions of the sleeve have a same diameter when the first longitudinal portion is not elastically stretched.
- (132) For some applications, the first and the second longitudinal portions of the sleeve are woven, and the first longitudinal portion of the sleeve is more loosely woven than the second longitudinal portion of the sleeve.
- (133) For some applications, the first longitudinal portion of the sleeve is radially stretchable, and the second longitudinal portion of the sleeve is substantially radially non-extensible.
- (134) For some applications, the annuloplasty ring further includes a plurality of tissue anchors, at least two of which are coupled to the sleeve at respective, different longitudinal sites alongside the elongated radial-force application member.
- (135) For some applications, the annuloplasty ring further includes a contracting assembly, which includes a contracting mechanism and a longitudinal contracting member, and the contracting

mechanism is fixed to the sleeve within 1.5 cm of the second sleeve end, measured when the sleeve is fully longitudinally extended.

(136) For some applications, the second coupling element is coupled to the contracting mechanism.

(137) For some applications, the longitudinal contracting member includes at least one wire.

(138) For some applications, the elongated radial-force application member includes metal.

(139) For some applications, the metal includes Nitinol.

(140) For some applications, at least one of the first and second coupling elements includes a hook.

(141) For some applications, at least one of the first and second coupling elements includes a loop.

(142) There is still further provided, in accordance with an application of the present invention, a method including: providing an annuloplasty ring, which includes (a) a flexible sleeve and (b) an elongated radial-force application element, which is disposed entirely within a longitudinal portion of the sleeve; during a percutaneous transcatheter procedure, placing the flexible sleeve entirely around an annulus of an atrioventricular valve of a subject, such that the longitudinal portion of the sleeve is disposed along an anterior portion of the annulus between fibrous trigones of the valve; and fastening the flexible sleeve at least to a posterior portion of the annulus, such that the elongated radial-force application element applies a force against the wall of the longitudinal portion of the sleeve in at least one radially-outward direction.

(143) For some applications, the elongated radial-force application element is rotationally asymmetric and not helically symmetric.

(144) For some applications, the elongated radial-force application element is configured to apply the force against the wall around less than 100% of a perimeter of the wall.

(145) For some applications, the elongated radial-force application element is configured to apply the force against the wall around less than 50% of the perimeter of the wall.

(146) For some applications, the elongated radial-force application element is configured to apply the force with a variation of less than 20% along a length of the elongated radial-force application element.

(147) For some applications, the flexible sleeve has first and second sleeve ends, and placing the flexible sleeve includes introducing the flexible sleeve into a left atrium while the first and the second sleeve ends are not coupled to each other; and thereafter, in the left atrium, arranging the flexible sleeve entirely around the annulus to form the closed loop.

(148) For some applications: the annuloplasty ring further includes (a) a first coupling element, which is coupled to the annuloplasty ring within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended, (b) a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the annuloplasty ring within 1.5 cm of the second sleeve end, measured when the sleeve is fully longitudinally extended, and coupling the first and the second sleeve ends to each other to form the closed loop includes coupling the first and the second coupling elements together.

(149) For some applications, the elongated radial-force application element has (a) a first radial-force-application-element longitudinal end that is between 2 and 6 cm from the first sleeve end, measured when the sleeve is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end that is within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended,

(150) For some applications, providing the annuloplasty ring includes providing the annuloplasty ring in which the sleeve is shaped so as to define an integrally closed loop having no sleeve ends.

(151) For some applications: the annuloplasty ring further includes a contracting assembly, which includes a housing that is fixed to the sleeve, and the elongated radial-force application element has (a) a first radial-force-application-element longitudinal end that is between 2 and 6 cm from the housing, measured when the sleeve is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end that is within 1.5 cm of the housing, measured when the sleeve is fully longitudinally extended.

(152) For some applications, the elongated radial-force application element includes an inflatable element.

(153) For some applications, placing the elongated radial-force application element includes placing the elongated radial-force application element along the anterior portion of the annulus, such that the elongated radial-force application element pushes the longitudinal portion of the sleeve against atrial tissue.

(154) For some applications, the annuloplasty ring further includes a substantially longitudinally non-extensible linking member, which has first and second linking-member ends and is at least partially disposed within the longitudinal portion of the sleeve, and the second linking-member end includes the first coupling element.

(155) For some applications, the linking member has a length of between 2 and 6 cm.

(156) For some applications, at least the longitudinal portion of the sleeve is substantially longitudinally non-extensible, and the first coupling element is fixed to the wall of the sleeve within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended.

(157) For some applications, the elongated radial-force application element includes a springy element.

(158) For some applications, placing the longitudinal portion of the sleeve includes twisting the longitudinal portion of the sleeve after fastening the sleeve to the portion of the annulus.

(159) For some applications, placing the longitudinal portion of the sleeve includes twisting the springy element after fastening the sleeve to the portion of the annulus.

(160) For some applications, where at least a portion of the springy element is curved at least partially about an inner surface of the wall of the sleeve.

(161) For some applications, at least a portion of the springy element is serpentine.

(162) For some applications, the at least a portion of the springy element is curved at least partially about the inner surface of the wall in a single circumferential direction.

(163) For some applications, at least a first portion of the springy element is curved at least partially about the inner surface of the wall in a first circumferential direction, and at least a second portion of the springy element is curved at least partially about the inner surface of the wall in a second circumferential direction circumferentially opposite the first circumferential direction.

(164) For some applications, at least a portion of the springy element is serpentine.

(165) For some applications, springy element includes a coiled spring.

(166) For some applications, the longitudinal portion of the sleeve is a first longitudinal portion, and the method further includes, after fastening the flexible sleeve at least to a posterior portion of the annulus, contracting a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve.

(167) For some applications, the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, and the annuloplasty ring further includes a longitudinal contracting member, which is arranged only along a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve.

(168) For some applications, the annuloplasty ring further includes a contracting assembly, which includes the longitudinal contracting member and a contracting mechanism.

(169) For some applications, a first average internal diameter of the first longitudinal portion of the sleeve is greater than a second average internal diameter of the second longitudinal portion of the sleeve, when both the first and the second longitudinal portions are fully radially expanded.

(170) For some applications, the first longitudinal portion of the sleeve is radially elastic, and the second longitudinal portion of the sleeve is substantially radially non-extensible.

(171) For some applications, the first and the second longitudinal portions of the sleeve are substantially longitudinally non-extensible.

(172) For some applications, the first and the second longitudinal portions of the sleeve have a same diameter when the first longitudinal portion is not elastically stretched.

- (173) For some applications, the first and the second longitudinal portions of the sleeve are woven, and the first longitudinal portion of the sleeve is more loosely woven than the second longitudinal portion of the sleeve.
- (174) For some applications, the first longitudinal portion of the sleeve is radially stretchable, and the second longitudinal portion of the sleeve is substantially radially non-extensible.
- (175) For some applications, the annuloplasty ring further includes a contracting assembly, which includes a contracting mechanism and a longitudinal contracting member, and the contracting mechanism is fixed to the sleeve within 30 mm of the second sleeve end, measured when the sleeve is fully longitudinally extended.
- (176) For some applications, the second coupling element is coupled to the contracting mechanism.
- (177) For some applications, the longitudinal contracting member includes at least one wire.
- (178) For some applications, the springy member includes metal.
- (179) For some applications, the metal includes Nitinol.
- (180) For some applications, at least one of the first and second coupling elements includes a hook.
- (181) For some applications, at least one of the first and second coupling elements includes a loop.
- (182) There is additionally provided, in accordance with an application of the present invention, apparatus including an implantable structure, which includes: a flexible sleeve, having first and second sleeve ends; a contracting assembly; an elongated linking member, having a first and second linking member ends, which second linking member end includes a first coupling element, wherein the linking member is coupled to the sleeve such that (a) at least a portion of the linking member is disposed within the sleeve, and (b) the first linking member end is longitudinally between the second linking member end and the first sleeve end, exclusive; and a second coupling element, which is configured to be coupleable to the first coupling element, and which is coupled to the implantable structure within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended.
- (183) For some applications, the contracting assembly is configured to longitudinal contract the sleeve.
- (184) For some applications, the implantable structure further includes a plurality of tissue anchors, at least two of which are coupled to the sleeve at respective, different longitudinal sites alongside the linking member.
- (185) For some applications, the contracting assembly includes a contracting mechanism and a longitudinal contracting member, and the contracting mechanism is coupled to the sleeve within 1.5 cm of the first sleeve end.
- (186) For some applications, the second coupling element is coupled to the contracting mechanism.
- (187) For some applications, the longitudinal contracting member includes at least one wire.
- (188) For some applications, the linking member is configured as a spring.
- (189) For some applications, the linking member is curved.
- (190) For some applications, the linking member has a length of between 2 and 6 cm.
- (191) For some applications, the linking member includes metal.
- (192) For some applications, the metal includes Nitinol.
- (193) For some applications, the linking member is substantially longitudinally non-extensible.
- (194) For some applications, at least 30% of a length of the linking member is disposed within the sleeve.
- (195) For some applications, at least 75% of the length of the linking member is disposed within the sleeve.
- (196) For some applications, the flexible sleeve is a first flexible sleeve, the implantable structure further includes a second flexible sleeve, and at least 20% of a length of the linking member is disposed within the second flexible sleeve.
- (197) For some applications, at least one of the first and second coupling elements includes a hook.
- (198) For some applications, at least one of the first and second coupling elements includes a loop.

(199) For some applications, the at least a portion of the linking member is disposed within a longitudinal portion of the sleeve, and the implantable structure further includes an elongated springy element, which is disposed within the longitudinal portion of the sleeve, and which is configured to apply a force against a wall of the longitudinal portion of the sleeve in at least one radially-outward direction.

(200) For some applications, the linking member is not configured as a spring.

(201) For some applications, the linking member is configured not to apply any force to the wall of the longitudinal portion of the sleeve.

(202) For some applications, at least 90% of a length of the linking member is straight when in a resting state.

(203) For some applications, the linking member is substantially longitudinally non-extensible.

(204) For some applications, the springy element has a length of between 2 and 6 cm, measured when the sleeve is fully longitudinally extended.

(205) For some applications: the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, the contracting assembly includes (a) a contracting mechanism, and (b) a longitudinal contracting member, which is arranged only along a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve, and the springy element is disposed entirely within the first longitudinal portion of the sleeve.

(206) For some applications, the first and the second coupling elements are configured to provide an adjustable-length connection between the linking member and the first sleeve end.

(207) There is yet additionally provided, in accordance with an application of the present invention, a method including: providing an implantable structure, which includes (a) a flexible sleeve, having first and second sleeve ends, (b) a contracting assembly, (c) an elongated linking member, having a first and second linking member ends, which second linking member end includes a first coupling element, wherein the linking member is coupled to the sleeve such that (i) at least a portion of the linking member is disposed within the sleeve, and (ii) the first linking member end is longitudinally between the second linking member end and the first sleeve end, exclusive, and (d) a second coupling element, which is coupled to the implantable structure within 1.5 cm of the first sleeve end, measured when the sleeve is fully longitudinally extended; during a percutaneous transcatheter procedure, placing the flexible sleeve around a portion of an annulus of an atrioventricular valve of a subject, which portion includes a posterior portion of the annulus; placing the linking member along an anterior portion of the annulus between fibrous trigones of the valve; fastening the flexible sleeve to the portion of the annulus; coupling the first and the second coupling elements together; and contracting a longitudinal portion of the sleeve.

(208) For some applications, the contracting assembly further includes a locking mechanism, and the method further includes, after contracting the longitudinal portion of the sleeve, locking the longitudinal contracting member with respect to the contracting assembly using the locking mechanism.

(209) For some applications, contracting the second longitudinal portion of the sleeve includes actuating the contracting assembly to contract the longitudinal portion of the sleeve.

(210) For some applications, fastening includes fastening the sleeve to the annulus using a plurality of tissue anchors, including coupling at least two of the anchors to the sleeve and tissue of the annulus at respective, different longitudinal sites alongside the linking member.

(211) For some applications, the contracting assembly includes a contracting mechanism and a longitudinal contracting member, and the contracting mechanism is coupled to the sleeve within 1.5 cm of the first sleeve end.

(212) For some applications, the second coupling element is coupled to the contracting mechanism.

(213) For some applications, the linking member is configured as a spring.

(214) For some applications, the linking member is curved.

(215) For some applications, the linking member has a length of between 2 and 6 cm.

(216) For some applications, the linking member includes metal.

(217) For some applications, the metal includes Nitinol.

(218) For some applications, the linking member is substantially longitudinally non-extensible.

(219) For some applications, at least 30% of a length of the linking member is disposed within the sleeve.

(220) For some applications, at least 75% of the length of the linking member is disposed within the sleeve.

(221) For some applications, the flexible sleeve is a first flexible sleeve, the implantable structure further includes a second flexible sleeve, and at least 20% of a length of the linking member is disposed within the second flexible sleeve.

(222) For some applications, at least one of the first and second coupling elements includes a hook.

(223) For some applications, at least one of the first and second coupling elements includes a loop.

(224) For some applications: the at least a portion of the linking member is disposed within a longitudinal portion of the sleeve, the implantable structure further includes an elongated springy element, which is disposed within the longitudinal portion of the sleeve, and placing the linking member includes placing the springy element along the anterior portion of the annulus, such that the springy element applies a force against a wall of the longitudinal portion of the sleeve in at least one radially-outward direction.

(225) For some applications, the linking member is not configured as a spring.

(226) For some applications, placing the linking member includes placing the linking member such that the linking member does not apply any force to the wall of the longitudinal portion of the sleeve.

(227) For some applications, at least 90% of a length of the linking member is straight when in a resting state.

(228) For some applications, the linking member is substantially longitudinally non-extensible.

(229) For some applications, the springy element has a length of between 2 and 6 cm, measured when the sleeve is fully longitudinally extended.

(230) For some applications: the longitudinal portion of the sleeve is a first longitudinal portion of the sleeve, the contracting assembly includes (a) a contracting mechanism, and (b) a longitudinal contracting member, which is arranged only along a second longitudinal portion of the sleeve that is entirely longitudinally distinct from the first longitudinal portion of the sleeve, and the springy element is disposed entirely within the first longitudinal portion of the sleeve.

(231) For some applications, the first and the second coupling elements are configured to provide an adjustable-length connection between the linking member and the first sleeve end, and placing the linking member along the anterior portion of the annulus includes setting an effective length of the linking member while coupling the first and the second coupling elements together.

(232) There is also provided, in accordance with an application of the present invention, apparatus including an annuloplasty system, which includes: an implantable structure, which includes a flexible sleeve, having first and second sleeve ends; a linking bridge element, which includes first and second bridge coupling interfaces, which are configured to be coupled to the sleeve in order to link the first and the second sleeve ends via the linking bridge element; and first and second flexible longitudinal guide members, which (a) are removably coupled to the sleeve within 1.5 cm of the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended, and (b) extend from the first and the second sleeve ends, respectively, away from the sleeve, and (c) removably pass through respective openings defined by the linking bridge member, so as to guide the first and the second bridge coupling interfaces to corresponding locations on the sleeve.

(233) For some applications, the respective openings defined by the linking bridge member are defined by the first and the second bridge coupling interfaces, respectively.

(234) For some applications, the sleeve includes first and second sleeve coupling interfaces, to

which the first and the second bridge coupling interfaces are configured to be coupled, respectively. (235) For some applications, the first and the second sleeve coupling interfaces are disposed within 1.5 cm of the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended.

(236) For some applications, the linking bridge element has a length of between 1 and 5 cm.

(237) For some applications, the implantable structure includes a longitudinal contracting member, which is configured to longitudinally contract a longitudinal portion of the sleeve, and the first and the second flexible longitudinal guide members are separate and distinct from the longitudinal contracting member.

(238) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, the first and the second flexible longitudinal guide members do not longitudinally overlap the longitudinal contracting member.

(239) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, no portion of either the first flexible longitudinal guide member or the second flexible longitudinal guide member is disposed more than 1.5 cm from the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended.

(240) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, the first and the second flexible longitudinal guide members are collectively disposed along less than 30% of a length of the sleeve, measured when the sleeve is fully longitudinally extended.

(241) There is further provided, in accordance with an application of the present invention, a method including: during a percutaneous transcatheter procedure, placing a flexible sleeve of an implantable structure partially around an annulus of a mitral valve of a subject, such that first and second flexible longitudinal guide members, which are removably coupled to the sleeve, extend from first and second sleeve ends of the sleeve, respectively, away from the sleeve, wherein the longitudinal guide members are removably coupled to the sleeve within 1.5 cm of the first and the second sleeve ends of the sleeve, respectively, measured when the sleeve is fully longitudinally extended; advancing a linking bridge element into a left atrium of the subject, while the longitudinal guide members removably pass through respective openings defined by the linking bridge member; using the first and the second longitudinal guide members to guide first and second bridge coupling interfaces of the linking bridge member to corresponding locations on the sleeve; and coupling the linking bridge member to the sleeve by coupling the first and the second bridge coupling interfaces to the sleeve, in order to link the first and the second sleeve ends via the linking bridge element.

(242) For some applications, the respective openings defined by the linking bridge member are defined by the first and the second bridge coupling interfaces, respectively.

(243) For some applications, the sleeve includes first and second sleeve coupling interfaces, and coupling the first and the second bridge coupling interfaces to the sleeve includes coupling the first and the second bridge coupling interfaces to the sleeve to the first and the second sleeve coupling interfaces, respectively.

(244) For some applications, the first and the second sleeve coupling interfaces are disposed within 1.5 cm of the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended.

(245) For some applications, the linking bridge element has a length of between 1 and 5 cm.

(246) For some applications: the implantable structure includes a longitudinal contracting member, the first and the second flexible longitudinal guide members are separate and distinct from the longitudinal contracting member, and the method further includes, after coupling the linking bridge member to the sleeve, contracting a longitudinal portion of the sleeve by causing the longitudinal contracting member to apply a contracting force to the longitudinal portion of the sleeve.

(247) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, the first and the second flexible longitudinal guide members do not longitudinally overlap the longitudinal contracting member.

(248) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, no portion of either the first flexible longitudinal guide member or the second flexible longitudinal guide member is disposed more than 1.5 cm from the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended.

(249) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, the first and the second flexible longitudinal guide members are collectively disposed along less than 30% of a length of the sleeve, measured when the sleeve is fully longitudinally extended.

(250) There is still further provided, in accordance with an application of the present invention, apparatus including an annuloplasty system, which includes: an implantable structure, which includes a flexible sleeve, having first and second sleeve ends; and first and second flexible longitudinal guide members, which (a) are removably coupled to the sleeve within 1.5 cm of the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended, and (b) extend from the first and the second sleeve ends, respectively, away from the sleeve.

(251) For some applications, the implantable structure includes a longitudinal contracting member, which is configured to longitudinally contract a longitudinal portion of the sleeve, and the first and the second flexible longitudinal guide members are separate and distinct from the longitudinal contracting member.

(252) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, the first and the second flexible longitudinal guide members do not longitudinally overlap the longitudinal contracting member.

(253) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, no portion of either the first flexible longitudinal guide member or the second flexible longitudinal guide member is disposed more than 1.5 cm from the first and the second sleeve ends, respectively, measured when the sleeve is fully longitudinally extended.

(254) For some applications, wherein, when the first and the second flexible longitudinal guide members are removably coupled to the sleeve, the first and the second flexible longitudinal guide members are collectively disposed along less than 30% of a length of the sleeve, measured when the sleeve is fully longitudinally extended.

(255) The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of a system for repairing a dilated atrioventricular valve, such as a mitral valve, in accordance with an application of the present invention;

(2) FIGS. 2A-I are schematic illustrations of a procedure for implanting the implantable structure of FIG. 1 to repair a mitral valve, in accordance with an application of the present invention;

(3) FIG. 3 is a schematic illustration of another configuration of the implantable structure of FIG. 1, prior to implantation, in accordance with an application of the present invention;

(4) FIG. 4 is a schematic illustration of the implantable structure of FIG. 3 after implantation around the annulus of a mitral valve, in accordance with an application of the present invention;

(5) FIG. 5 is a schematic illustration of a closed-loop configuration of the implantable structure of FIG. 1, in accordance with an application of the present invention;

(6) FIG. 6 is a schematic illustration of yet another configuration of the implantable structure of FIG. 1, prior to implantation, in accordance with an application of the present invention;

(7) FIGS. 7A-B are schematic illustrations of the implantable structure of FIG. 6 after implantation around the annulus of a mitral valve, in accordance with respective applications of the present invention;

(8) FIGS. 8A-D are schematic illustrations of coupling elements, in accordance with respective applications of the present invention;

(9) FIG. 9 is a schematic illustration of another configuration of the implantable structure of FIG. 1, prior to implantation, further comprising an elongated radial-force application element, in accordance with an application of the present invention;

(10) FIG. 10 is a schematic illustration of the implantable structure of FIG. 9 implanted around the mitral valve, in accordance with an application of the present invention;

(11) FIGS. 11A-D are schematic illustrations of several configurations of the elongated radial-force application element of the implantable structure of FIGS. 9 and 10, in accordance with an application of the present invention;

(12) FIG. 12 is a schematic illustration of another configuration of the elongated radial-force application element of the implantable structure of FIGS. 9 and 10, in accordance with an application of the present invention;

(13) FIG. 13 is a schematic illustration of yet another configuration of the elongated radial-force application element of the implantable structure of FIGS. 9 and 10, in which the elongated radial-force application element is helically symmetric, in accordance with an application of the present invention;

(14) FIG. 14 is a schematic illustration of a configuration of the sleeve of the implantable structure of FIGS. 9 and 10, in accordance with an application of the present invention;

(15) FIG. 15 is a schematic illustration of another configuration of the implantable structure of FIGS. 9 and 10, in accordance with an application of the present invention;

(16) FIGS. 16A-B are schematic illustrations of another configuration of the implantable structure of FIGS. 9 and 10, in which the sleeve is shaped so as to define an integrally closed loop having no sleeve ends, in accordance with an application of the present invention;

(17) FIG. 17 is a schematic illustration of another configuration of the implantable structure of FIG. 9 implanted around the mitral valve, in accordance with an application of the present invention;

(18) FIGS. 18A and 18B are schematic illustrations of yet another configuration of the implantable structure of FIG. 1, prior to implantation and upon implantation around the mitral valve, respectively, in accordance with an application of the present invention;

(19) FIG. 19 is a schematic illustration of still another configuration of the implantable structure of FIG. 1 implanted around the mitral valve, in accordance with an application of the present invention;

(20) FIG. 20 is a schematic illustration of another configuration of the implantable structure of FIG. 1 implanted around the mitral valve, in accordance with an application of the present invention;

(21) FIG. 21 is a schematic illustration of still another configuration of the implantable structure of FIG. 1 implanted around the mitral valve, in accordance with an application of the present invention;

(22) FIGS. 22A-D are schematic illustrations of another system for repairing a dilated atrioventricular valve, and a method for deploying the system, in accordance with an application of the present invention;

(23) FIGS. 23A-B are schematic illustrations of another configuration of a linking bridge element of the system of FIGS. 22A-D, in accordance with an application of the present invention;

(24) FIG. 24 is a schematic illustration of a contracting mechanism, disassembled to show a

relationship among individual components of the contracting mechanism, in accordance with an application of the present invention; and

(25) FIGS. **25A-B** and **26** are schematic illustrations of a valve prosthesis assembly, in accordance with respective applications of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

(26) FIG. **1** is a schematic illustration of a system **20** for repairing a dilated atrioventricular valve, such as a mitral valve or a tricuspid valve, in accordance with an application of the present invention. System **20** comprises an adjustable implantable structure **22**, shown in FIG. **1** in a straight, relaxed, non-contracted state, and an anchor deployment manipulator **24** (shown in FIGS. **2G-H**). For some applications, implantable structure **22** is configured to be deployed as an annuloplasty ring, while for other applications, implantable structure **22** is configured to be deployed as a base ring to which a prosthetic valve is coupled, such as described hereinbelow with reference to FIG. **25A-B** or **26**. Implantable structure **22** comprises a flexible sleeve **26**. Anchor deployment manipulator **24** is advanced into sleeve **26**, as shown in FIGS. **2G-H**, and, from within the sleeve, deploys tissue anchors through a wall of the sleeve into cardiac tissue, thereby anchoring the ring around at least a portion of the valve annulus. For some applications, anchor deployment manipulator is implemented using techniques described in US Patent Application Publication 2010/0280604, which is incorporated herein by reference, with reference to FIGS. **2, 3, 4, 5A, 5B, 6A, 6B, 7, 8, 13**, and/or **20A-E** thereof.

(27) For some applications, implantable structure **22** comprises a partial annuloplasty ring. In these applications, sleeve **26** is configured to be placed only partially around the valve annulus (i.e., to assume a C-shape), and, once anchored in place, to be contracted so as to circumferentially tighten the valve annulus. For other applications, sleeve **26** is configured to be implanted entirely around the valve annulus in a closed loop, such as described hereinbelow with reference to FIG. **4, 5, 7A-B, 10, 16A-B, 17** or **18B**.

(28) Implantable structure **22** further comprises a contracting assembly **40**, which facilitates contracting of the implantable structure. Contracting assembly **40** typically comprises a contracting mechanism **28**, and a longitudinal contracting member **30**, which is coupled to contracting mechanism **28**, extends along a portion of the sleeve, and is typically flexible. For example, contracting member **30** may comprise at least one wire. Contracting assembly **40** is configured to contract a longitudinal portion of sleeve **26**, and is described in more detail hereinbelow. In addition, the implantable structure typically comprises a plurality of tissue anchors **38**, typically between about 5 and about 20 anchors, such as about 10 or about 16 anchors. In FIG. **1**, anchors **38** are shown coupled to implantable structure **22**, deployed through the wall of sleeve **26**. For some applications, anchors **38** are configured as described with reference to FIGS. **5A-C, 5D, 5E, 5F, 5G, 5H**, and/or **5I** in US Patent Application Publication 2012/0330411, which is incorporated herein by reference, while for other applications, anchors **38** comprise tissue anchors known in the art.

(29) Flexible sleeve **26** may comprise a braided, knitted, or woven mesh or a tubular structure comprising ePTFE. For some applications, the braid comprises metal and fabric fibers. The metal fibers, which may comprise Nitinol for example, may help define the shape of the sleeve, e.g., hold the sleeve open to provide space for passage and manipulation of deployment manipulator **24** within the sleeve. The fabric fibers may promote tissue growth into the braid. Typically, sleeve **26** is substantially longitudinally non-extensible, i.e., a length thereof is substantially constant, i.e., cannot be longitudinally stretched, under normal usage conditions. Alternatively, the sleeve is somewhat elastic, which gives the sleeve a tendency to longitudinally contract, thereby helping tighten the sleeve. For example, the sleeve may be bellows- or accordion-shaped.

(30) For some applications, the sleeve is configured to have a tendency to assume a straight shape when in its relaxed, non-contracted state. This straightness may help the surgeon locate the next site for each subsequent anchor during the implantation procedure. For example, because the sleeve assumes a generally straight shape, the sleeve may help provide an indication of distance between

adjacent anchoring sites. For some applications, the sleeve is configured to have a controllably variable stiffness. For example, a somewhat stiff wire may be placed in the sleeve to provide the stiffness, and subsequently be removed at the conclusion of the implantation procedure when the stiffness is no longer useful.

(31) For some applications, sleeve **26** comprises a plurality of radiopaque markers **39**, which are positioned along the sleeve at respective longitudinal sites. The markers may provide an indication in a radiographic image (such as a fluoroscopy image) of how much of the sleeve has been deployed at any given point during an implantation procedure, in order to enable setting a desired distance between anchors **38** along the sleeve. For some applications, the markers comprise a radiopaque ink.

(32) Typically, at least a portion (e.g., at least three, such as all) of the longitudinal sites are longitudinally spaced at a constant interval. Typically, the longitudinal distance between the distal edges of adjacent markers, and/or the distance between the proximal edges of adjacent markers, is set equal to the desired distance between adjacent anchors. For example, the markers may comprise first, second, and third markers, which first and second markers are adjacent, and which second and third markers are adjacent, and the distance between the proximal and/or distal edges of the first and second markers equal the corresponding distance between the proximal and/or distal edges of the second and third markers. For example, the distance may be between 3 and 15 mm, such as 6 mm, and the longitudinal length of each marker may be between 0.1 and 14 mm, such as 2 mm. (If, for example, the distance were 6 mm and the length were 2 mm, the longitudinal gaps between adjacent markers would have lengths of 4 mm.)

(33) Longitudinal contracting member **30** comprises a wire, a ribbon, a rope, or a band, which typically comprises a flexible and/or superelastic material, e.g., nitinol, polyester, HDPE, stainless steel, or cobalt chrome. For some applications, the wire comprises a radiopaque material. For some applications, longitudinal contracting member **30** comprises a braided polyester suture (e.g., Ticron). For some applications, longitudinal contracting member **30** is coated with polytetrafluoroethylene (PTFE). For some applications, contracting member **30** comprises a plurality of wires that are intertwined to form a rope structure. For some applications, implantable structure **22** comprises a plurality of contracting members **30**, which may extend along generally the same longitudinal portion of sleeve **26**, or along respective, different portions of sleeve **26** (e.g., as described with reference to FIG. **13** in above-mentioned US Patent Application Publication 2012/0330411).

(34) For some applications, contracting member **30** is positioned at least partially within a lumen of the sleeve **26**, such as entirely within the lumen (as shown in FIGS. **1**, **2H-I**, **3**, **4**, **6**, and **7A-B**). For some applications in which the contracting member is positioned partially within the lumen, the contracting member is sewn into the wall of the sleeve, such that the contracting member is alternatingly inside and outside of the sleeve along the length of the sleeve (configuration not shown). Optionally, sleeve **26** defines an internal channel within which member **30** is positioned (configuration not shown). Alternatively, the contracting member is disposed outside the lumen of the sleeve, such as alongside an outer wall of the sleeve. For example, sleeve **26** may define an external channel within which contracting member **30** is positioned, or the sleeve may comprise or be shaped so as to define external coupling elements, such as loops or rings (configuration not shown). For some applications, contracting member **30** is positioned approximately opposite the anchors.

(35) For some applications of the present invention, contracting mechanism **28** comprises a rotatable structure, such as a spool **46**. The rotatable structure is arranged such that rotation thereof applies a longitudinal contracting force, thereby contracting at least a longitudinal portion of implantable structure **22**. Typically, in these applications, contracting mechanism **28** further comprises a housing **44** in which the rotatable structure, e.g., the spool, is positioned. Contracting member **30** has first and second member ends, and a first member end portion, which extends from

the first member end toward the second member end and along only a longitudinal portion of the contracting member. For some applications, the first member end portion, e.g., the first member end of contracting member **30**, is coupled to contracting mechanism **28**, such as the rotatable structure, e.g., the spool (alternatively, although the first member end portion is coupled to the contracting mechanism, the first member end protrudes beyond the contracting mechanism). For example, spool **46** may be shaped to provide a hole **42** or other coupling mechanism for coupling the first end of contracting member **30** to the spool, and thereby to contracting mechanism **28**. Contracting assembly **40** is arranged such that rotation of the spool winds a portion of the contracting member around the spool. Alternatively, contracting member **30** may comprise at least one wire (e.g., exactly one wire) that passes through a coupling mechanism of spool **46**, in order to couple the wire to the spool. The ends of the wire are brought together, and together serve as a second end **53** of contracting member **30**. In this configuration, approximately the longitudinal center of the wire serves as the first end of the contracting member.

(36) Alternatively, contracting mechanism **28** may comprise a ratchet contracting mechanism, which typically comprises a ratchet-coupling housing. Contracting member **30** is shaped so as to define engaging structures, such as grooves or teeth. Techniques may be used that are described in International Application PCT/IL2009/000593, filed Jun. 15, 2009, which published as PCT Publication WO 10/004546, and in U.S. application Ser. No. 12/996,954, which published as US Patent Application Publication 2011/0166649, in the national stage thereof, all of which applications and publications are incorporated herein by reference.

(37) Further alternatively, contracting mechanism **28** may comprise a housing or other structure (e.g., a ring or an eyelet) which is shaped so as to define an opening therethrough. Contracting member **30** is drawn through the opening (such that the first member end protrudes beyond the opening), and, once a desired length has been achieved, is locked, such as using a locking bead, or by crimping or knotting.

(38) Contracting member **30** extends along less than the entire length of sleeve **26**. Contracting mechanism **28** (e.g., housing **44** thereof) is disposed at a first site **34** of sleeve **26** that is a first longitudinal distance **D1** from a first end of the sleeve, either a proximal end **49** of sleeve **26**, as shown in FIG. **1**, or a distal end **51** of sleeve **26**, as shown in FIGS. **2G-I**. (Longitudinal distance **D1** is measured between the first end of the sleeve and the portion of contracting mechanism **28** that is closest to the first end.) For some applications, second end **53** of contracting member **30** is coupled to the sleeve at a second site **36** that is a second longitudinal distance **D2** from a second end of the sleeve, which second end is longitudinally opposite the first end of the sleeve. For applications in which contracting mechanism **28** comprises a rotatable structure, rotation of the rotatable structure, such as spool **46**, longitudinally contracts at least a portion of the sleeve, such as by winding a portion of the contracting member around the spool, thereby pulling the far end of the implantable structure toward the spool and shortening and tightening the implantable structure. Such rotation of the rotatable structure, or other actuation of contracting assembly **40**, typically applies a longitudinal contracting force only between first and second sites **34** and **36**, which longitudinally contracts at least a portion, e.g. all, of the sleeve only between first and second sites **34** and **36**. (For example, the longitudinal force may longitudinally contract less than the entire sleeve between first and second sites **34** and **36** in applications in which system **20** comprises coiled element **240**, which provides a contraction-restricting portion of the sleeve, as described hereinbelow with reference to FIGS. **10A-E** and/or **11A-E** in above-mentioned US Patent Application Publication 2012/0330411.) Therefore, the portions of the sleeve beyond first and second sites **34** and **36** (towards the ends of the sleeve) are not contracted by contracting assembly **40**.

(39) Typically, contracting member **30** extends along (i.e., a distance along the sleeve between first and second sites **34** and **36** equals) no more than 80% of the length of the sleeve, e.g., no more than 60% or no more than 50% of the length. Typically, contracting member **30** extends along no more

than 80% of a circumference of the loop when the sleeve is placed around the annulus (i.e., the total length of the loop less the length of any overlapping portion). Typically, contracting member **30** extends along (i.e., a distance along the sleeve between first and second sites **34** and **36** equals) at least 20% of the length of the sleeve, e.g., at least 40% or at least 50% of the length. Typically, contracting member **30** extends along at least 20% of the circumference of the loop when the sleeve is placed around the annulus, e.g., at least 30% or at least 50%.

(40) For some applications, first longitudinal distance **D1**, measured when sleeve **26** is in a straight, relaxed, non-contracted state, is at least 3 mm, e.g., at least 5 mm, such as at least 9 mm, e.g., at least 14 mm; no greater than 20 mm, such as no greater than 15 mm; and/or between 5 and 20 mm, such as between 9 and 15 mm. Alternatively or additionally, for some applications, second longitudinal distance **D2**, measured when sleeve **26** is in a straight, relaxed, non-contracted state, is at least 3 mm, e.g., at least 5 mm, such as at least 9 mm, e.g., at least 14 mm; no greater than 20 mm, such as no greater than 15 mm; and/or between 5 and 20 mm, such as between 9 and 15 mm. Further alternatively or additionally, first longitudinal distance **D1**, measured when sleeve **26** is in a straight, relaxed, non-contracted state, is no greater than 20%, such as no greater than 10% of a total length of the sleeve, measured when sleeve **26** is in a straight, relaxed, non-contracted state. Further alternatively or additionally, second longitudinal distance **D2**, measured when sleeve **26** is in a straight, relaxed, non-contracted state, is no greater than 30%, such as no greater than 20%, e.g., no greater than 10% of the total length of the sleeve measured, when sleeve **26** is in a straight, relaxed, non-contracted state. For some applications, the total length of the sleeve, measured when the sleeve is in a straight, relaxed, non-contracted state is at least 5 cm, no more than 25 cm, and/or between 5 and 25 cm. For some applications in which the sleeve is implanted in a closed loop, the total length of the sleeve is selected to be between 1.3 and 1.4 times a circumference of the annulus, in order to provide overlapping portion **114**, described hereinbelow with reference to FIGS. **3** and **4**.

(41) For some applications, at least one of tissue anchors **38** (e.g., exactly one, at least two, exactly two, at least three, exactly three, or at least four, or no more than four) is coupled to sleeve **26** longitudinally between contracting mechanism **28** (e.g., housing **44** thereof) and the first sleeve end (i.e., the end of the sleeve to which contracting mechanism **28** is closest), exclusive, and at least 3, such as at least 6, of tissue anchors **38** are coupled to the sleeve alongside contracting member **30**, longitudinally between first site **34** and second site **36** (second member end **53**), exclusive. (As used in the present application, including in the claims, “exclusive,” when used with respect to a range of locations, means excluding the endpoints of the range.)

(42) Alternatively or additionally, for some applications, at least one of tissue anchors **38** (e.g., exactly one, at least two, exactly two, at least three, exactly three, or at least four, or no more than four) is coupled to sleeve **26** longitudinally between second site **36** (second member end **53**) and the second sleeve end (i.e., the end of the sleeve to which second member end **53** is closest), exclusive, and at least 3, such as at least 6, of tissue anchors **38** are coupled to the sleeve alongside contracting member **30**, longitudinally between first site **34** and second site **36** (second member end **53**), exclusive.

(43) In the exemplary configuration shown in FIG. **1**, exactly two tissue anchors **38** are coupled to the sleeve longitudinally between the contracting mechanism (e.g., the housing) (first site **34**) and the first sleeve end, exclusive, exactly two tissue anchors are coupled to the sleeve longitudinally between first site **34** and second site **36** (second member end **53**), exclusive, and exactly six tissue anchors **38** are coupled to the sleeve alongside the contracting member, longitudinally between first site **34** and second site **36** (second member end **53**), exclusive.

(44) Providing the one or more anchors beyond first and second sites **34** and **36** (i.e., beyond the contracting portion of contracting member **30**) generally distributes force applied by contraction of contracting assembly **40** over these anchors. In contrast, in some configurations of implantable structure **22** in which anchors are not provided beyond first and second sites **34** and **36**, the force

applied by the contracting assembly is applied predominantly to the single anchor nearest the first end of the contracting member, and the single anchor nearest to second end of the contracting member.

(45) For some applications, anchors **38** are positioned along sleeve **26** with a longitudinal distance of between 4.5 and 9 mm, such as 6 mm, between each pair of longitudinally-adjacent anchors.

(46) It is noted that the anchors may be positioned as described above by a surgeon during an implantation procedure, such as described hereinbelow with reference to FIGS. 2A-I, or the anchors may be prepositioned in the sleeve.

(47) Reference is now made to FIGS. 2A-I, which are schematic illustrations of a procedure for implanting implantable structure **22** to repair a mitral valve **130**, in accordance with an application of the present invention. The procedure is typically performed with the aid of imaging, such as fluoroscopy, transesophageal echo, and/or echocardiography.

(48) The procedure typically begins by advancing a semi-rigid guidewire **102** into a right atrium **120** of the patient, as shown in FIG. 2A.

(49) As shown in FIG. 2B, guidewire **102** provides a guide for the subsequent advancement of a sheath **104** therealong and into the right atrium. Once sheath **104** has entered the right atrium, guidewire **102** is retracted from the patient's body. Sheath **104** typically comprises a 14-20 F sheath, although the size may be selected as appropriate for a given patient. Sheath **104** is advanced through vasculature into the right atrium using a suitable point of origin typically determined for a given patient. For example: sheath **104** may be introduced into the femoral vein of the patient, through an inferior vena cava **122**, into right atrium **120**, and into a left atrium **124** transseptally, typically through the fossa ovalis; sheath **104** may be introduced into the basilic vein, through the subclavian vein to the superior vena cava, into right atrium **120**, and into left atrium **124** transseptally, typically through the fossa ovalis; or sheath **104** may be introduced into the external jugular vein, through the subclavian vein to the superior vena cava, into right atrium **120**, and into left atrium **124** transseptally, typically through the fossa ovalis.

(50) For some applications, sheath **104** is advanced through an inferior vena cava **122** of the patient (as shown) and into right atrium **120** using a suitable point of origin typically determined for a given patient.

(51) Sheath **104** is advanced distally until the sheath reaches the interatrial septum.

(52) As shown in FIG. 2D, a resilient needle **106** and a dilator (not shown) are advanced through sheath **104** and into the heart. In order to advance sheath **104** transseptally into left atrium **124**, the dilator is advanced to the septum, and needle **106** is pushed from within the dilator and is allowed to puncture the septum to create an opening that facilitates passage of the dilator and subsequently sheath **104** therethrough and into left atrium **124**. The dilator is passed through the hole in the septum created by the needle. Typically, the dilator is shaped to define a hollow shaft for passage along needle **106**, and the hollow shaft is shaped to define a tapered distal end. This tapered distal end is first advanced through the hole created by needle **106**. The hole is enlarged when the gradually increasing diameter of the distal end of the dilator is pushed through the hole in the septum.

(53) The advancement of sheath **104** through the septum and into the left atrium is followed by the extraction of the dilator and needle **106** from within sheath **104**, as shown in FIG. 2E.

(54) As shown in FIG. 2F, implantable structure **22** (with anchor deployment manipulator **24** therein) is advanced through sheath **104** into left atrium **124**.

(55) As shown in FIG. 2G, distal end **51** of sleeve **26** is positioned in a vicinity of a left fibrous trigone **142** of an annulus **140** of mitral valve **130**. (It is noted that for clarity of illustration, distal end **51** of sleeve **26** is shown schematically in the cross-sectional view of the heart, although left fibrous trigone **142** is in reality not located in the shown cross-sectional plane, but rather out of the page closer to the viewer.) Alternatively, the distal end is positioned in a vicinity of a right fibrous trigone **144** of the mitral valve (configuration not shown). Further alternatively, the distal end of the

sleeve is not positioned in the vicinity of either of the trigones, but is instead positioned elsewhere in a vicinity of the mitral valve, such as in a vicinity of the anterior or posterior commissure. Still further alternatively, for some applications, the distal end is positioned along an anterior portion of the annulus, such as described hereinbelow with reference to FIG. 4. For some applications, outer tube **66** of anchor deployment manipulator **24** is steerable, as is known in the catheter art, while for other applications, a separate steerable tube is provided, such as described in the above-mentioned '604 publication, with reference to FIG. 15 and FIG. 16 thereof. In either case, the steering functionality typically allows the area near the distal end of the deployment manipulator to be positioned with six degrees of freedom. Once positioned at the desired site near the selected trigone, deployment manipulator **24** deploys a first anchor **38** through the wall of sleeve **26** into cardiac tissue near the trigone.

(56) As shown in FIG. 2H, deployment manipulator **24** is repositioned along annulus **140** to another site selected for deployment of a second anchor **38**. Typically, the first anchor is deployed most distally in the sleeve (generally at or within a few millimeters of the distal end of the sleeve), and each subsequent anchor is deployed more proximally, such that the sleeve is gradually pulled off (i.e., withdrawn from) the deployment manipulator in a distal direction during the anchoring procedure. The already-deployed first anchor **38** holds the anchored end of sleeve **26** in place, so that the sleeve is drawn from the site of the first anchor towards the site of the second anchor. Typically, as the sleeve is pulled off (i.e., withdrawn from) the deployment manipulator, the deployment manipulator is moved generally laterally along the cardiac tissue, as shown in FIG. 2H. Deployment manipulator **24** deploys the second anchor through the wall of the sleeve into cardiac tissue at the second site. Depending on the tension applied between the first and second anchor sites, the portion of sleeve **26** therebetween may remain tubular in shape, or may become flattened, which may help reduce any interference of the implantable structure with blood flow.

(57) For some applications, in order to provide the second and subsequent anchors, anchor driver **68** is withdrawn from the subject's body via sheath **104** (typically while leaving outer tube **66** of the deployment manipulator in place in the sleeve), provided with an additional anchor, and then reintroduced into the subject's body and into the outer tube. Alternatively, the entire deployment manipulator, including the anchor driver, is removed from the body and subsequently reintroduced upon being provided with another anchor. Further alternatively, deployment manipulator **24** is configured to simultaneously hold a plurality of anchors, and to deploy them one at a time at the selected sites.

(58) As shown in FIG. 2I, the deployment manipulator is repositioned along the annulus to additional sites, at which respective anchors are deployed, until the last anchor is deployed in a vicinity of right fibrous trigone **144** (or left fibrous trigone **142** if the anchoring began at the right trigone), thereby fastening sleeve **26** and implantable structure **22** to the annulus. Alternatively, the last anchor is not deployed in the vicinity of a trigone, but is instead deployed elsewhere in a vicinity of the mitral valve, such as in a vicinity of the anterior or posterior commissure.

(59) For applications in which contracting mechanism **28** comprises spool **46**, a rotation tool is typically used to rotate spool **46** of contracting mechanism **28**, in order to tighten implantable structure **22**. For some applications, the rotation tool is used that is described and shown in the above-mentioned '604 publication, with reference to FIGS. 6A-B, 7, and 8 thereof. As described therein, contracting mechanism **28** comprises longitudinal member **86** that is attached to the contracting mechanism and passes out of the body of the subject, typically via sheath **104**. In order to readily bring the rotation tool to a driving interface of contracting mechanism **28**, the rotation tool is guided over longitudinal member **86**. For some applications, spool **46** is configured as described in the '604 publication with reference to FIGS. 1-4, 6A-B, 7, and/or 8 thereof.

(60) Contracting assembly **40** typically comprises a locking mechanism that locks contracting member **30** with respect to contracting assembly **40**, thereby preventing loosening (and typically tightening) of contracting member **30**. For some applications, spool **46** comprises the locking

mechanism that prevents rotation of the spool after contracting member **30** has been tightened. For example, locking techniques may be used that are described and shown in US Application Publication 2010/0161047, which is incorporated herein by reference, with reference to FIG. **4** thereof, and/or with reference to FIGS. **6B**, **7**, and **8** of the above-mentioned '604 publication. Alternatively, for some applications, contracting mechanism **28** is configured to tighten contracting member **30**, crimp the contracting member to hold the contracting member taut, and subsequently cut the excess length of the contracting member.

(61) For some applications, a rotation handle is used to tighten the implantable structure, such as described and shown in the above-mentioned '604 publication, with reference to FIGS. **9A-C** and **10A-D** thereof. As mentioned above, deploying the one or more anchors beyond the contracting portion of contracting member **30** generally distributes force applied by contraction of contracting assembly **40** over these anchors.

(62) For some applications, sleeve **26** is filled with a material (e.g., polyester, polytetrafluoroethylene (PTFE), polyethylene terephthalate (PET), or expanded polytetrafluoroethylene (ePTFE)) after being implanted. The material is packed within at least a portion, e.g., 50%, 75%, or 100%, of the lumen of sleeve **26**. The filler material functions to prevent (1) formation within the lumen of sleeve **26** of clots or (2) introduction of foreign material into the lumen which could obstruct the sliding movement of contracting member **30**.

(63) For some applications, proximal end **49** of sleeve **26** is closed upon completion of the implantation procedure. Alternatively, the proximal end of the sleeve may have a natural tendency to close when not held open by deployment manipulator **24**.

(64) For some applications, following initial contraction of implantable structure **22** during the implantation procedure, the structure may be further contracted or relaxed at a later time after the initial implantation, such as between several weeks and several months after the initial implantation. Using real-time monitoring and tactile feedback, optionally in combination with fluoroscopic imaging, a rotation tool or anchor driver of a deployment manipulator may be reintroduced into the heart and used to contract or relax implantable structure **22**.

(65) Reference is now made to FIGS. **3** and **4**, which are schematic illustrations of another configuration of implantable structure **22**, in accordance with an application of the present invention. FIG. **3** shows implantable structure **22** in a straight, relaxed, non-contracted state, prior to implantation. FIG. **4** shows the implantable structure after implantation around the annulus of mitral valve **130**, in accordance with an application of the present invention.

(66) In this configuration, sleeve **26** is implanted in a closed loop. More particularly, a first portion **110** of sleeve **26** longitudinally extends from the first sleeve end (i.e., the end of the sleeve to which contracting mechanism **28**, e.g., housing **44** thereof, is closest) toward contracting mechanism **28**, e.g., housing **44** thereof (but typically does not extend all of the way to the contracting mechanism), and a second portion **112** of the sleeve longitudinally extends from the second sleeve end (i.e., the end of the sleeve to which second member end **53** is closest) toward second member end **53** (but typically does not extend all of the way to the second member end). As shown in FIG. **4**, once implanted, sleeve **26** is arranged in a closed loop, such that first and second portions **110** and **112** of the sleeve together define a longitudinally overlapping portion **114** of the sleeve. The overlapping portion typically has a length of at least 2 mm (e.g., at least 5 mm), no more than 60 mm (e.g., no more than 50 mm), and/or between 2 mm (e.g., 5 mm) and 60 mm (e.g., 50 mm), and/or a length that is at least 1% of a total length of the sleeve, no more than 40% of the total length (e.g., no more than 30%), and/or between 1% and 40% (e.g., 30%) of the total length of the sleeve, measured when the sleeve is in a straight, relaxed, non-contracted state.

(67) For some applications, at least one of tissue anchors **38** (labeled as **38E** in FIGS. **3** and **4**) penetrates both first and second portions **110** and **112** of the sleeve at overlapping portion **114**. Such a mutual anchor helps ensure that the first and second portions remain tightly coupled together and to the tissue, so that the sleeve retains its closed loop shape. For some applications in which tissue

anchor **38E** comprises a coupling head and a tissue coupling element, such as described hereinbelow with reference to FIG. 5D, 5E, 5F, 5G, or **5I** in above-mentioned US Patent Application Publication 2012/0330411, the tissue coupling element penetrates both first and second portions **110** and **112** of the sleeve at overlapping portion **114**, and the coupling head is positioned within one of first and second portions **110** and **112** of the sleeve at the overlapping portion. For example, in the deployment configuration shown in FIG. 4, the coupling head of anchor **38E** is positioned within second portion **112**.

(68) This configuration of implantable structure **22** may be implanted using the procedure described hereinabove with reference to FIGS. 2A-I, with the following differences. Unlike in the deployment shown in FIGS. 2G-I, in this configuration sleeve **26** is deployed as a closed band around the entire annulus of the native valve, including an anterior portion **116** of the annulus (on the aortic side of the valve) between fibrous trigones **142** and **144**. Typically, both first and second portions **110** and **112** of sleeve **26** (and thus overlapping portion **114**) are positioned along anterior portion **116** of the annulus.

(69) For some applications, during the implantation procedure, the first sleeve end (i.e., the end of the sleeve to which contracting mechanism **28**, e.g., housing **44** thereof, is closest) is placed along at least a portion of anterior portion **116** and first portion **110** is extended along this portion. At least one anchor **38D** is deployed through the wall of first portion **110** of sleeve **26** into cardiac tissue at the anterior portion of the annulus. Additional anchors **38A** and/or **38C** are deployed through the wall of the sleeve around the non-anterior remainder of the annulus, including the posterior portion thereof, as described hereinabove with reference to FIG. 2H. (Anchors **38C**, if provided, are deployed beyond the ends of the contracting portion of contracting member **30**, while anchors **38A** are deployed along the portion of the sleeve including the contracting portion of the contracting member.)

(70) A portion of the sleeve is placed on at least a portion of anterior portion **116** of the annulus, and, typically, one or more anchors **38B** are deployed through the wall of the sleeve into tissue at the anterior portion of the annulus.

(71) The sleeve is further extended around the annulus until second portion **112** overlaps with previously-deployed first portion **110** at overlapping portion **114**, forming a complete ring. At least one anchor **38E** is deployed from within second portion **112** through the wall of the sleeve and into the cardiac tissue, typically at anterior portion **116** of the annulus, or at a portion of the annulus near anterior portion **116**. Typically, anchor **38E** is deployed such that it additionally passes through previously-deployed first portion **110** (passing through the wall of first portion **110** twice).

(Optionally, anchors **38B** and/or **38E** are of a different configuration than anchors **38A**, **38C**, and/or **38D**, such as described with reference to FIGS. 5A-I in above-mentioned US Patent Application Publication 2012/0330411; anchors **38B** and **38E** may be of the same configuration as one another, or of different configurations.)

(72) Alternatively, the second sleeve end (i.e., the end of the sleeve to which second member end **53** is closest) is first placed at least partially along anterior portion **116**, in which case second portion **112** is deployed before first portion **110**, and anchor **38E** is deployed from within first portion **110**.

(73) The sleeve may be deployed in either a clockwise direction or a counterclockwise direction, as viewed from the atrium.

(74) Contracting assembly **40** is actuated, e.g., the rotatable structure of contracting mechanism **28** is rotated, in order to tighten implantable structure **22**, as described hereinabove with reference to FIG. 2I. Typically, contracting member **30** does not extend along the portion of sleeve **26** deployed along anterior portion **116** of the annulus, and thus does not extend along first portion **110**, second portion **112**, or overlapping portion **114** of sleeve **26**. The portion of the sleeve deployed along anterior portion **116** of the annulus (between the trigones) is thus non-contractible. For some applications, contracting member **30** is positioned along a non-anterior portion of the annulus,

which non-anterior portion does not reach either of the fibrous trigones, e.g., does not reach within 5 mm of either of the trigones. Tightening of implantable structure **22** therefore tightens at least a portion of the posterior portion of the annulus, while preserving the length of anterior portion **116** of the annulus. (The anterior portion of the annulus should generally not be contracted because its tissue is part of the skeleton of the heart.) However, the portion of the sleeve deployed along the anterior portion of the annulus prevents dilation of the anterior annulus, because the sleeve is anchored at both ends of the anterior annulus, and, as mentioned above, the sleeve typically comprises a longitudinally non-extensible material. This deployment configuration may help prevent long-term resizing of the anterior annulus, which sometimes occurs after implantation of partial annuloplasty rings, such as C-bands.

(75) For some applications, the non-contractible portion of sleeve **26** (the portion without contracting member **30**) extends somewhat beyond one or both of trigones **142** or **144** (in the posterior direction, away from anterior portion **116** of the annulus), such as up to 20 mm, such as up to 10 mm. In general, since the non-contractible portions of the sleeve are preset, the surgeon is able to decide during the implantation procedure the lengths of the anterior non-contractible area and the posterior contractible area, by selecting the length of overlapping portion **114**. The greater the length of overlapping portion **114**, the greater the relative length of the posterior contractible portion, and the lesser the relative length of the non-contractible portion.

(76) For some applications, at least one anchor **38C** is coupled to cardiac tissue on the posterior side of right fibrous trigone **144**, between the trigone and the end of contracting member **30**. Similarly, at least one anchor **38C** may be coupled to cardiac tissue on the posterior side of left fibrous trigone **142**, between the trigone and the other end of contracting member **30** (which, for some applications, is coupled to contracting mechanism **28**, as shown in FIG. **4**).

(77) For some applications, at least one (either one or both) of first and second longitudinal distances **D1** and **D2** (described hereinabove with reference to FIG. **1**), taken separately, is greater than 40 mm, such as greater than 60 mm. This sleeve portion(s) beyond the contracting portion of contracting member **30** provide the non-contractible portion of the sleeve positioned along anterior portion **116** of the annulus, and, optionally, the non-contractible portion(s) that extend beyond the anterior portion.

(78) Reference is still made to FIGS. **3** and **4**. For some applications, anchors **38** deployed along anterior portion **116** of the annulus (between the trigones) are of a different configuration from anchors **38** deployed along the remainder of the annulus (including the posterior portion of the annulus). Unlike the remainder of the annulus, anterior portion **116** does not comprise muscular or fibrous tissue, but rather thinner aortic tissue (typically the anchors positioned along anterior portion **116** enter the aorta below the aortic leaflets). The anchors that are deployed along the remainder of the annulus are configured for strong coupling to the thicker and stronger fibrous tissue of these portions of the annulus. Such anchors may be inappropriate for coupling to anterior portion **116**. Anchors **38** are thus provided that are particularly configured for coupling to anterior portion **116**. For example, different configurations of anchors **38** are described with reference to FIGS. **5A-I** in above-mentioned US Patent Application Publication 2012/0330411.

(79) For these applications, anchors **38** include a plurality of first tissue anchors of a first configuration, and a plurality of second tissue anchors of a second configuration different from the first configuration. (The first tissue anchors are labeled **38A** and **38C** in FIG. **4**, and for the sake of brevity, are referenced as **38A** hereinbelow. The second tissue anchors are labeled **38B**, **38D**, and **38E** in FIG. **4**, and for the sake of brevity, are referenced as **38B** hereinbelow.) For some applications, implantable structure **22** comprises more first tissue anchors **38A** than second tissue anchors **38B**, e.g., at least twice as many first tissue anchors as second tissue anchors.

(80) For these applications, sleeve **26** is typically arranged as a loop. For example, as described hereinabove with reference to FIG. **4**, the sleeve may be shaped so as to define first and second sleeve ends, which are coupled to each other (optionally, with overlapping portion **114**) to form the

loop. Alternatively, as described hereinbelow with reference to FIG. 6, the sleeve may be shaped so as to define an integrally closed loop having no sleeve ends. First tissue anchors **38A** are coupled to sleeve **26** at intervals along a first longitudinally-contiguous portion of the loop, and second tissue anchors **38B** are coupled to sleeve **26** at intervals along a second longitudinally-contiguous portion of the loop different from the first longitudinally-contiguous portion. The second portion of the loop is deployed along anterior portion **116** of the annulus, and the first portion of the loop is deployed along at least a portion of the remainder of the annulus (including the posterior portion of the annulus).

(81) Reference is made to FIG. 5, which is a schematic illustration of an alternative closed-loop configuration of implantable structure **22**, in accordance with an application of the present invention. In this configuration, flexible sleeve **26** is shaped so as to define an integrally closed loop having no sleeve ends. For some applications, anchors **38** deployed along anterior portion **116** of the annulus are of a different configuration from anchors **38** deployed along the remainder of the annulus, as described hereinabove with reference to FIGS. 3-4. The anchors may be configured as described with reference to FIGS. 5A-I in above-mentioned US Patent Application Publication 2012/0330411.

(82) Typically, contracting member **30** does not extend along the portion of sleeve **26** deployed along anterior portion **116** of the annulus. The portion of the sleeve deployed along anterior portion **116** of the annulus (between the trigones) is thus non-contractible. Tightening of implantable structure **22** therefore tightens at least a portion of the posterior portion of the annulus, while preserving the length of anterior portion **116** of the annulus. (The anterior portion of the annulus should generally not be contracted because its tissue is part of the skeleton of the heart.) However, the portion of the sleeve deployed along the anterior portion of the annulus prevents dilation of the anterior annulus, because the sleeve is anchored at both ends of the anterior annulus, and, as mentioned above, the sleeve typically comprises a longitudinally non-extensible material. This deployment configuration may help prevent long-term resizing of the anterior annulus, which sometimes occurs after implantation of partial annuloplasty rings, such as C-bands.

(83) For some applications, the non-contractible portion of sleeve **26** (the portion without contracting member **30**) extends somewhat beyond one or both of trigones **142** or **144** (in the posterior direction, away from anterior portion **116** of the annulus), such as up to 20 mm, such as up to 10 mm.

(84) For some applications, at least one anchor **38** is coupled to cardiac tissue on the posterior side of right fibrous trigone **144**, between the trigone and the end of contracting member **30**. Similarly, at least one anchor **38** may be coupled to cardiac tissue on the posterior side of left fibrous trigone **142**, between the trigone and the other end of contracting member **30** (which, for some applications, is coupled to contracting mechanism **28**, as shown in FIG. 5).

(85) Reference is now made to FIGS. 6 and 7A-B, which are schematic illustrations of another configuration of implantable structure **22**, in accordance with an application of the present invention. FIG. 6 shows implantable structure **22** in a relaxed, non-contracted state, and FIGS. 7A-B shows the implantable structure implanted around mitral valve **130**. This configuration of implantable structure **22** is generally similar to the configuration described hereinabove with reference to FIG. 1, except as follows. In this configuration, implantable structure **22** further comprises an elongated linking member **250**, which is positioned at least partially along anterior portion **116** of the annulus, so as to join the ends of implantable structure **22** in a complete loop. Over time after implantation, linking member **250** becomes fixed to anterior portion **116** of the annulus, thereby helping prevent long-term dilation of the anterior annulus. Typically, at least a portion (e.g., at least 30%, such as at least 75% or at least 90%) of a length of linking member **250** is disposed within and covered by sleeve **26**, into and/or over which fibrous tissue grows over time, helping anchor the linking member to tissue of the anterior annulus. Alternatively or additionally, a separate flexible sleeve or a coating (e.g., a polymeric coating) may be provided that covers at least

20%, e.g., between 20% and 80%, of the linking member. Typically, in the configuration of implantable structure **22** shown in FIGS. **6** and **7A-B**, none of anchors **38** is coupled to anterior portion **116** of the annulus. For some applications, as shown in FIG. **7A**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications, as shown in FIG. **7B**, implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**. This latter arrangement may facilitate placement of the first-deployed, distal-most anchor **38** near right fibrous trigone **144**, which is above the fossa ovalis, and the linking of first and second coupling elements **256** and **260** later in the implantation procedure.

(86) Linking member **250** has first and second linking member ends **252** and **254**. Second linking member end **254** comprises (e.g., is shaped so as to define, or is fixed to) a first coupling element **256**. First linking member end **252** is disposed longitudinally between second linking member end **254** and a first sleeve end (either proximal end **49**, as shown, or distal end **51**, not shown), exclusive. Second linking member end **254** either protrudes from the second end of the sleeve, or is recessed within the second end of the sleeve (as shown, the second end of the sleeve is distal end **51**). A longitudinal portion of linking member **250** in a vicinity of first linking member end **252** is coupled to the sleeve. For example, the portion may be threaded through the fabric of the sleeve, and/or sewn (e.g., sutured) to the fabric of the sleeve to hold the linking member in place during deployment, and the linking member may be held in place after implantation by one or more of anchors **38**, such as two or more anchors **38F**. Optionally, the linking member is not initially coupled to the sleeve, but is instead held in place by a delivery tool during the implantation procedure, until being coupled to the sleeve by one or more of the anchors, for example. The coupled longitudinal portion may have a length of between 2 and 10 mm, and optionally includes first linking member end **252** of the linking member.

(87) Implantable structure **22** further comprises a second coupling element **260**, which is configured to be coupleable to first coupling element **256**. Second coupling element **260** typically is coupled to implantable structure **22** within 1.5 cm of the first end of sleeve **26** (opposite the end mentioned above near which first linking member end **252** is fixed), measured when the sleeve is fully longitudinally extended. As mentioned above, in the configuration shown in FIGS. **6** and **7A-B**, this first end is proximal end **49**.

(88) For some applications, such as shown in FIGS. **6** and **7A-B**, contracting mechanism **28** (e.g., housing **44** thereof) is disposed along sleeve **26** within 30 mm, such as within 15 mm, of the first sleeve end (i.e., the same end of the sleeve near which the second coupling element is coupled), measured when sleeve **26** is fully longitudinally extended. For example, contracting mechanism **28** (e.g., housing **44** thereof) may be fixed at the first sleeve end. Alternatively, for some applications, contracting mechanism **28** (e.g., housing **44** thereof) is fixed at least 5 mm from the first sleeve end, e.g., between 5 and 30 mm, such as between 5 and 15 mm, from the first sleeve end. Second coupling element **260** may be coupled to contracting mechanism **28** (e.g., to housing **44**).

Alternatively, second coupling element **260** may be otherwise coupled to sleeve **26** (such as directly coupled), in which case contracting mechanism **28**, e.g., housing **44** thereof, may be coupled to sleeve **26** at a greater longitudinal distance from the end of the sleeve, and one or more of anchors **38** may be coupled to the sleeve longitudinally between the contracting mechanism and the sleeve end, such as described hereinabove with reference to FIGS. **1**, **2A-I**, **3**, and **4**.

(89) Typically, linking member **250** is substantially longitudinally non-extensible, i.e., its length is fixed. Typically, linking member **250** comprises metal, such as Nitinol or stainless steel. For some applications, the linking member has a length of at least 2 cm, no more than 6 cm, and/or between 2 and 6 cm.

(90) For some applications, the linking member is configured as a spring, which is typically curved, so as to be elastic in a radial direction, i.e., to be compressible like a bow or deflected beam. In these applications, the linking member is oriented such that it is pressed by elasticity

against the anterior portion of the mitral annulus, i.e., the outer wall of the aorta, thereby holding the sleeve covering the linking member against the aortic wall.

(91) For some applications, at least two of tissue anchors **38** are coupled to sleeve **26** at respective, different longitudinal sites alongside linking member **250**, within 6 cm of first linking member end **252**, such as within 2 to 6 cm of the first end. These tissue anchors may help set the proper direction of curvature of the linking member, for applications in which the linking member is curved.

(92) Reference is made to FIGS. **8A-D**, which are schematic illustrations of coupling elements **256** and **260**, in accordance with respective applications of the present invention. For some applications, at least one of first and second coupling elements **256** and **260** comprises a hook **270**. Alternatively or additionally, for some applications, at least one of the first and second coupling elements comprises a loop **272**. In the configuration shown in FIG. **8A** (and FIGS. **6** and **7A-B**), first coupling element **256** comprises hook **270**, and second coupling element **260** comprises a loop **272**. In the configuration shown in FIG. **8B**, both first and second coupling elements **256** and **260** comprises respective loops **272**, and the coupling elements are coupled together such as by placing one of anchors **38** through both loops and into cardiac tissue.

(93) For some applications, first and second coupling elements **256** and **260** are configured to provide an adjustable-length connection between linking member **250** and the first end of sleeve. Such an adjustable-length connection allows the effective length of linking member **250** to be set during the implantation procedure in order to accommodate variations in individual patient anatomy. For some applications, such as shown in FIG. **8C**, first coupling element **256** comprises a flexible elongate member **274**, which comprises a plurality of protrusions **276** distributed along a portion of flexible elongate member **274**. Flexible elongate member **274** is drawn through a loop defined by second coupling element **260** until a desired length of linking member **250** is achieved; one of the protrusions prevents loosening. Alternatively, second coupling element **260** comprises flexible elongate member **274** (having protrusions **276**), and first coupling element **256** defines the loop through which flexible elongate member **274** is drawn (configuration not shown). For some applications, such as shown in FIG. **8D**, first coupling element **256** comprises a plurality of loops **272**, arranged longitudinally (each loop is connected to an adjacent loop, either directly or such as by a short length of wire), and second coupling element **260** comprises a single loop **272**. The healthcare professional selects which of loops **272** of first coupling element **256** to couple with the single loop **272** of second coupling element **260**, in order to set the length of linking member **250**. Alternatively, second coupling element **260** comprises the plurality of loops **272**, and first coupling element **256** comprises the single loop **272**, or both first and second coupling elements **256** and **260** comprise pluralities of loops (configurations not shown).

(94) Reference is now made to FIGS. **9-17**, which are schematic illustrations of additional configurations of implantable structure **22**, in accordance with respective applications of the present invention. FIGS. **9** and **12-16B** show implantable structure **22** (which typically comprises an annuloplasty ring) in a relaxed, non-contracted state. FIGS. **11A-D** show several configurations of an elongated radial-force application element **482**, labeled with reference numerals **482A**, **482B**, **482C**, and **482D**, respectively. These configurations of implantable structure **22** are generally similar to the configuration described hereinabove with reference to FIG. **1**, except as follows, and may incorporate any of the features of the configuration described hereinabove with reference to FIG. **1**, *mutatis mutandis*.

(95) FIG. **10** shows implantable structure **22** implanted around mitral valve **130**, before a longitudinal portion of sleeve **26** has been contracted. For some applications, as shown in FIG. **10**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**. This latter arrangement may facilitate placement of the first-deployed, distal-

most anchor **38** near right fibrous trigone **144**, which is above the fossa ovalis, and the linking of first and second coupling elements **456** and **260** later in the implantation procedure, for applications in which these coupling elements are provided, such as described hereinbelow.

(96) In these configurations, implantable structure **22** further comprises elongated radial-force application element **482**, which is disposed entirely within a first longitudinal portion of sleeve **26**. Elongated radial-force application element **482** is configured to apply a force against a wall of the first longitudinal portion of sleeve **26** in at least one radially-outward direction. The applied force pushes the first longitudinal portion of sleeve **26** against tissue of the left atrium, such as against tissue of the annulus and/or the atrial wall, so as to inhibit blood flow between sleeve **26** and the tissue. It is generally desirable to inhibit blood flow between sleeve **26** and the annulus on anterior side, to avoid creating turbulence.

(97) For some applications, elongated radial-force application element **482** is configured to apply a force against the wall of at least 20 gram-force, no more than 1 kg-force, and/or between 20 gram-force and 1 kg-force, such as at least 50 gram-force, no more than 500 gram-force (e.g., no more than 300 gram-force), and/or between 50 gram-force and 500 gram-force (e.g., between 50 gram-force and 300 gram-force). For some applications, elongated radial-force application element **482** is configured to apply the force generally constantly along the length of elongated radial-force application element **482**, e.g., with a variation of less than 20% along the length.

(98) When implanting implantable structure **22**, elongated radial-force application element **482** is placed along anterior portion **116** of the annulus, between fibrous trigones **142** and **144** (a portion of elongated radial-force application element **482** may extend beyond one or both of the trigones, such as for coupling to anchors **38F**, as described hereinbelow). If, upon initial placement, radial-force application element **482** does not apply the force against the wall of sleeve **26** in the desired radial direction (e.g., in the direction of the atrial wall), the healthcare professional may rotate the radial-force application element **482** within the sleeve, and/or rotate (e.g., twist) the first longitudinal portion of sleeve **26**. Typically, longitudinal portion **480** extends along at least 20 mm of anterior portion **116** of the annulus, and/or along at least 20%, no more than 100%, and/or between 20% and 100% of anterior portion **116** of the annulus, such as at least 30%, no more than 60%, and/or between 30% and 60% of anterior portion **116**. Typically, in the configuration of implantable structure **22** shown in FIGS. **9-10** and **12-16B**, none of anchors **38** is coupled to anterior portion **116** of the annulus.

(99) Typically, elongated radial-force application element **482** has a length of no more than 6 cm, measured when sleeve **26** is fully longitudinally extended.

(100) For some applications, elongated radial-force application element **482** is rotationally asymmetric and not helically symmetric, such as shown in FIGS. **9-12** and **14-16B**.

(101) For some applications, such as shown in FIGS. **9-16B**, elongated radial-force application element **482** comprises a springy element **484**. For some applications, at least a portion of springy element **484** is curved at least partially about an inner surface of the wall of sleeve **26**, such as shown in FIGS. **9, 10, 12, 14, 15**, and **16A-B**. Typically, springy element **484** comprises an elastic material, such as a metal, such as Nitinol or stainless steel.

(102) For some applications (such as when elongated radial-force application element **482** comprises springy element **484**), as labeled in FIGS. **11A-D**, elongated radial-force application element **482** is shaped so as to define one or more axial base sections **510** (e.g., exactly two axial base sections **510A** and **510B**, as shown in FIGS. **11A-D**), and one or more offset sections **512** (e.g., exactly one offset section **512**, as shown in FIG. **11A** (and FIG. **10**), or a plurality of offset sections **512** (e.g., between two and 20, e.g., between two and ten, such as between two and six), as shown in FIGS. **11B-D**). The one or more axial base sections **510** are coaxial with a longitudinal axis **514** of elongated radial-force application element **482**, and the one or more offset sections **512** are not coaxial with longitudinal axis **514**. A greatest distance **D** between the one or more offset sections **512** and longitudinal axis **514** is typically at least 2 mm, no more than 10 mm (e.g., no

more than 6 mm), and/or between 2 and 10 mm (e.g., between 2 and 6 mm), e.g., 4 mm.

(103) For some applications, offset section(s) **512** are at least partially straight, such as shown in FIGS. **11A** and **11B**. For some applications, offset sections **512** are at least partially curved, such as shown in FIGS. **11C** and **11D**. For some applications, offset sections **512** are at least partially serpentine, such as shown in FIG. **11D**.

(104) For some applications, the at least a portion of springy element **484** is curved at least partially about the inner surface of the wall of sleeve **26** in a single circumferential direction, such as shown in FIGS. **9**, **10**, **14**, **15**, and **16A-B**. Alternatively, for some applications, at least a first portion **485A** of springy element **484** is curved at least partially about the inner surface of the wall of sleeve **26** in a first circumferential direction **486A**, and at least a second portion **485B** of springy element **484** is curved at least partially about the inner surface of the wall of sleeve **26** in a second circumferential direction **486B** circumferentially opposite the first circumferential direction, such as shown in FIG. **12**.

(105) This configuration may use any of the shapes shown in FIGS. **11A-D** (with the shapes doubled), or other shapes. This configuration pushes against the wall of sleeve **26** and the tissue at at least two circumferential locations around the sleeve, and may help hold the rotational position of the sleeve, allow less accurate rotational alignment, and/or help compensate for anatomical variability.

(106) For some applications, such as shown in Section A-A of FIG. **9**, elongated radial-force application element **482** is configured to apply the force against the wall of sleeve **26** around an angle α (alpha) that is less than 100% of a perimeter of the wall of sleeve **26** around a central longitudinal axis **516** of sleeve **26**, such as around less than 75%, e.g., less than 50%, such as less than 25%, of the perimeter of the wall of sleeve **26**. (Central longitudinal axis **516** runs along sleeve **26**; the cross-section shown in Section A-A of FIG. **9** is perpendicular to the central longitudinal axis.) Force is not required to be applied around 100% of the perimeter of the wall of sleeve **26** because a circumferential portion of the wall faces the blood-filled volume of the chamber, rather than atrial tissue, and there would be no benefit to pushing the wall of sleeve **26** against the blood-filled volume.

(107) For some applications, such as shown in FIG. **13**, elongated radial-force application element **482** is helically symmetric; for these applications, springy element **484** typically comprises a coiled spring **490**. For some applications, when in a relaxed state, coiled spring **490** has an outer diameter of at least 2.5 mm, no more than 10 mm, and/or between 2.5 and 10 mm, such as at least 3.5 mm, no more than 6 mm, and/or between 3.5 and 6 mm. For some applications, when in a relaxed state, the outer diameter of coiled spring **490** is greater than (e.g., equals at least 110% of, such as at least 130% of, e.g. at least 150% of) an inner diameter of a second longitudinal portion **492** that is entirely longitudinally distinct from first longitudinal portion **480** of sleeve **26**, when second longitudinal portion **492** is fully radially expanded. Coiled spring **490** is typically initially held constrained with a smaller diameter in a separate tube smaller than the inner diameter of the deployment sheath.

(108) Reference is again made to FIGS. **9-16B**. For some applications, longitudinal contracting member **30** of contracting assembly **40** is arranged only along at least a portion of second longitudinal portion **492**. For some of these applications, contracting assembly **40** is configured to contract the at least a portion of the second longitudinal portion **492**.

(109) Reference is made to FIG. **14**. For some applications, first and second longitudinal portions **480** and **492** of sleeve **26** are configured such that first longitudinal portion **480** either has, or is configured to assume, a first average internal diameter **D1** that is greater than a second average internal diameter **D2** of second longitudinal portion **492**. For example, first average internal diameter **D1** may be at least 110% of **D2**, such as at least 150% of second average internal diameter **D2**. This larger average diameter enables elongated radial-force application element **482** (e.g., springy element **484**) to push a large surface area of sleeve **26** against the atrial tissue, thereby

better encouraging tissue growth, better inhibiting blood between the sleeve and the atrial tissue, and accommodating variations in individual patient anatomy. For some applications, first and second longitudinal portions **480** and **492** collectively extend along an entire length of sleeve **26**. This configuration, as well as the various options described below, may be used in combination with any of the configurations described herein with reference to FIGS. **9-13** and/or **15**.

(110) For some applications, first average internal diameter **D1** of first longitudinal portion **480** of sleeve **26** is greater than second average internal diameter **D2** of second longitudinal portion **492** of sleeve **26**, when both first and second longitudinal portions **480** and **492** are fully radially expanded (in these applications, typically both first and second longitudinal portions **480** and **492** are substantially radially non-extensible).

(111) For some other applications, first longitudinal portion **480** of sleeve **26** is radially elastic and thus able to stretch from an initial smaller average internal diameter to first average internal diameter **D1**, and second longitudinal portion **492** of sleeve **26** is substantially radially non-extensible, and thus cannot expand to a diameter beyond second average internal diameter **D2**. For example, first longitudinal portion **480** may comprise a first plurality of substantially non-extensible fibers that extend longitudinally along the first longitudinal portion, and a second plurality of elastic fibers that are arranged circumferentially around the first longitudinal portion (typically, woven with the first plurality of fibers). Typically, first and second longitudinal portions **480** and **492** of sleeve **26** are substantially longitudinally non-extensible, i.e., a length thereof is substantially constant, i.e., cannot be longitudinally stretched, under normal usage conditions. Optionally, first and second longitudinal portions **480** and **492** of sleeve **26** have a same diameter (equal to second average internal diameter **D2**) when first longitudinal portion **480** is not elastically stretched. Alternatively, for some applications, first and second longitudinal portions **480** and **492** of sleeve **26** are woven, and first longitudinal portion **480** of sleeve **26** is more loosely woven than second longitudinal portion **492** of sleeve **26**. Further alternatively, for some applications, first longitudinal portion **480** of sleeve **26** is radially stretchable, and second longitudinal portion **492** of sleeve **26** is substantially radially non-extensible. For example, first longitudinal portion **480** may comprise a first plurality of substantially non-extensible fibers that extend longitudinally along the first longitudinal portion, and a second plurality of stretchable fibers that are arranged circumferentially around the first longitudinal portion (typically, woven with the first plurality of fibers).

(112) For some applications, such as shown in FIGS. **9-10** and **12-15**, sleeve **26** has (a) a first sleeve end **51** (which may correspond to distal end **51** of sleeve **26**, as shown, or to the proximal end, configuration not shown), and (b) a second sleeve end **49** (which may correspond to proximal end **49** of sleeve **26**, as shown, or to the distal end, configuration not shown). For some applications, elongated radial-force application element **482** has (a) a first radial-force-application-element longitudinal end **496** that is between 2 and 6 cm from first sleeve end **51**, measured when sleeve **26** is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end **498** that is within 1.5 cm of first sleeve end **51**, measured when sleeve **26** is fully longitudinally extended.

(113) For some applications, such as shown in FIGS. **9-10** and **12-15**, implantable structure **22** (e.g., the annuloplasty ring) further comprises (a) a first coupling element **456**, which is coupled to the annuloplasty ring within 1.5 cm of first sleeve end **51**, measured when sleeve **26** is fully longitudinally extended, and (b) second coupling element **260**, as described above with reference to FIGS. **6** and **7A-B**. Second coupling element **260** is configured to be coupleable to first coupling element **456**, and is fixed to implantable structure **22** (e.g., the annuloplasty ring) within 1.5 cm of second sleeve end **49**, measured when sleeve **26** is fully longitudinally extended. For some applications, at least one of first and second coupling elements **456** and **260** comprises a hook. Alternatively or additionally, for some applications, at least one of the first and second coupling elements comprises a loop. For example, in the configurations shown in FIGS. **9-15**, first coupling

element **456** comprises a hook, and second coupling element **260** comprises a loop. Alternatively, for example, both the first and the second coupling elements comprises loops, such as shown in FIGS. **8B** and **8D**, and the coupling elements are coupled together such as by placing one of anchors **38** through both loops and into cardiac tissue.

(114) Elongated radial-force application element **482** is typically fixed to sleeve **26** at least near first radial-force-application-element longitudinal end **496**, such that elongated radial-force application element **482** is arranged as a cantilever. Typically, elongated radial-force application element **482** is fixed to sleeve **26** at least near first radial-force-application-element longitudinal end **496**, such that first radial-force-application-element longitudinal end **496** is rotationally fixed with respect to the sleeve, in order to allow twisting of elongated radial-force application element **482** to store spring energy in elongated radial-force application element **482** near first radial-force-application-element longitudinal end **496**. The shape of first radial-force-application-element longitudinal end **496** may aid in rotationally fixing the end with respect to the sleeve. For example, first radial-force-application-element longitudinal end **496** may include a circumferentially-oriented component, as shown in the figures.

(115) A portion of elongated radial-force application element **482** may be threaded through the fabric of the sleeve, and/or sewn (e.g., sutured) to the fabric of the sleeve to hold the elongated radial-force application element in place during deployment, and/or the elongated radial-force application element may be held in place after implantation by one or more of anchors **38**, such as two or more anchors **38F**.

(116) For some applications, such as shown in FIGS. **9-15**, contracting mechanism **28** (e.g., housing **44** thereof) is fixed along sleeve **26** within 30 mm, such as within 15 mm, of second sleeve end **49** (i.e., the same end of the sleeve near which second coupling element **260** is coupled), measured when sleeve **26** is fully longitudinally extended. For example, contracting mechanism **28** (e.g., housing **44** thereof) may be fixed at second sleeve end **49**.

(117) Alternatively, for some applications, contracting mechanism **28** (e.g., housing **44** thereof) is fixed at least 5 mm from second sleeve end **49**, e.g., between 5 and 30 mm, such as between 5 and 15 mm, from second sleeve end **49**. Second coupling element **260** may be coupled to contracting mechanism **28** (e.g., to housing **44**). Alternatively, second coupling element **260** may be otherwise coupled to sleeve **26** (such as directly coupled), in which case contracting mechanism **28**, e.g., housing **44** thereof, may be coupled to sleeve **26** at a greater longitudinal distance from the end of the sleeve, and one or more of anchors **38** may be coupled to the sleeve longitudinally between the contracting mechanism and the sleeve end, such as described hereinabove with reference to FIGS. **1**, **2A-I**, **3**, and **4**.

(118) For some applications, such as shown in FIGS. **9-14**, implantable structure **22** (e.g., the annuloplasty ring) further comprises a substantially longitudinally non-extensible linking member **450**, i.e., a length thereof is substantially constant, i.e., cannot be longitudinally stretched, under normal usage conditions. Linking member **450** typically helps prevent long-term dilation of the anterior annulus. Linking member **450** is typically configured not to apply any force to the wall of first longitudinal portion **480** of sleeve **26**.

(119) Typically, linking member **450** is not configured as a spring. For some applications, linking member **450** comprises a metal (e.g., Nitinol or stainless steel) or a polymer. For some applications, linking member **450** is rigid, while for other applications, the linking member is not rigid.

(120) Linking member **450** has first and second linking-member ends **452** and **454**. Linking member **450** is at least partially disposed within and covered by first longitudinal portion **480** of sleeve **26**. Typically, at least 30%, such as at least 75% or at least 90% of a length of linking member **450** is disposed within and covered by first longitudinal portion **480** of sleeve **26**. Over time after implantation, linking member **450** becomes fixed to anterior portion **116** of the annulus. Second linking-member end **454** comprises (e.g., is shaped so as to define, or is fixed to) first coupling element **456**. Second linking-member end **454** either protrudes from first sleeve end **51**, or

is recessed within first sleeve end **51**. A longitudinal portion of linking member **450** in a vicinity of first linking-member end **452** is typically coupled to sleeve **26**. For example, the portion may be threaded through the fabric of the sleeve, and/or sewn (e.g., sutured) to the fabric of the sleeve to hold the linking member in place during deployment. Optionally, a longitudinal portion of linking member **450** in a vicinity of first linking-member end **452** is held in place after implantation by one or more of anchors **38**, such as two or more anchors **38F** (configuration not shown). Optionally, the linking member is not initially coupled to the sleeve, but is instead held in place by a delivery tool during the implantation procedure, until being coupled to the sleeve during the implantation procedure. Typically, linking member **250** has a length of at least 2 cm, no more than 6 cm, and/or between 2 and 6 cm.

(121) For some applications, at least first longitudinal portion **480** of sleeve **26** is substantially longitudinally non-extensible, i.e., a length thereof is substantially constant, i.e., cannot be longitudinally stretched, under normal usage conditions. In these applications, first longitudinal portion **480** typically helps prevent long-term dilation of the anterior annulus.

(122) For some applications, such as shown in FIG. **15**, first coupling element **456** is fixed to the wall of sleeve **26** within 1.5 cm of first sleeve end **51**, measured when sleeve **26** is fully longitudinally extended. Implantable structure **22** typically does not comprise linking member **450** in these applications. In these applications, at least first longitudinal portion **480** of sleeve **26** is substantially longitudinally non-extensible, and first longitudinal portion **480** typically helps prevent long-term dilation of the anterior annulus.

(123) Reference is made to FIGS. **9-15**. Typically, sleeve **26** is placed entirely around an annulus of the atrioventricular valve, e.g., the mitral valve. For applications in which sleeve **26** has first and second sleeve ends **51** and **49**, as described hereinabove with reference to FIGS. **9-14**, sleeve **26** is introduced into the left atrium while first and second sleeve ends **51** and **49** are not coupled to each other, and thereafter, in the left atrium, sleeve **26** is arranged entirely around the annulus to form the closed loop.

(124) Reference is still made to FIGS. **9-16**. For some applications, during placement, after fastening sleeve **26** to the portion of the annulus, the healthcare professional twists elongated radial-force application element **482** (and optionally first longitudinal portion **480** of sleeve **26**), and then, typically, links first and second coupling elements **456** and **260**. Optionally, such twisting may serve one or both of the following purposes: (1) the twisting may store energy in springy element **484** for exertion of torque against the wall of the sleeve, and (2) the twisting may rotationally align springy element **484** in the desired radial direction. Alternatively or additionally to twisting for the first of these purposes, springy element **484** may be pre-loaded (twisted) to store energy before implantation in the subject, such as immediately before implantation or during manufacture.

(125) Reference is again made to FIGS. **8A-D**. The techniques described with reference to these figures regarding coupling element **256** may be implemented for coupling element **456** of the configuration described with reference to FIGS. **9-15**.

(126) Reference is made to FIGS. **16A-B**, which are schematic illustrations of implantable structure **22** in which sleeve **26** is shaped so as to define an integrally closed loop having no sleeve ends, in accordance with respective applications of the present invention. In these applications, the wall of sleeve **26** typically is shaped so as to define a lateral opening **500** through which anchor deployment manipulator **24** is introduced. For some applications, elongated radial-force application element **482** has (a) a first radial-force-application-element longitudinal end **496** that is at least 2 cm, no more than 6 cm, and/or between 2 and 6 cm from housing **44** of contracting assembly **40** (housing **44** is fixed to sleeve **26**), measured when sleeve **26** is fully longitudinally extended, and (b) a second radial-force-application-element longitudinal end **498** that is within 1.5 cm of housing **44**, measured when sleeve **26** is fully longitudinally extended. Alternatively, for some applications, first radial-force-application-element longitudinal end **496** is within 1.5 cm of housing **44**,

measured when sleeve **26** is fully longitudinally extended, and second radial-force-application-element longitudinal end **498** is at least 2 cm, no more than 6 cm, and/or between 2 and 6 cm from housing **44** of contracting assembly **40**, measured when sleeve **26** is fully longitudinally extended. (127) For some applications, such as shown in FIG. **16B**, first and second longitudinal portions **480** and **492** of sleeve **26** are configured such that first longitudinal portion **480** either has, or is configured to assume, a first average internal diameter **D1** that is greater than a second average internal diameter **D2** of second longitudinal portion **492**. For example, first average internal diameter **D1** may be at least 110% of second average internal diameter **D2**, such as at least 150% of second average internal diameter **D2**. First average internal diameter **D1** may be achieved using the techniques described hereinabove with reference to FIG. **14**. For other applications, such as shown in FIG. **16A**, the entire sleeve (i.e., first and second longitudinal portions **480** and **492**) has a constant internal diameter.

(128) For some applications, as shown in FIGS. **16A-B**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**.

(129) Reference is now made to FIG. **17**, which is a schematic illustration of another configuration of implantable structure **22** implanted around the mitral valve, in accordance with an application of the present invention. In this configuration, elongated radial-force application element **482** comprises an inflatable element **494**, such as a balloon. After fastening sleeve **26** to the portion of the annulus (and, optionally, after linking first and second coupling elements **456** and **260**), the healthcare professional inflates inflatable element **494**, typically with a liquid (such as saline solution) or a gel. For some applications, inflatable element **494** is provided separately from implantable structure **22**, and the healthcare professional introduces inflatable element **494**, while uninflated, into sleeve **26**, typically after fastening sleeve **26** to the portion of the annulus (and, optionally, after linking first and second coupling elements **456** and **260**), and then inflates inflatable element **494**. These inflation techniques may be used with any of the techniques described herein with reference to FIGS. **9-16B**, mutatis mutandis.

(130) For some applications, as shown in FIG. **17**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**. This latter arrangement may facilitate placement of the first-deployed, distal-most anchor **38** near right fibrous trigone **144**, which is above the fossa ovalis, and the linking of first and second coupling elements **456** and **260** later in the implantation procedure, for applications in which these coupling elements are provided, such as described hereinbelow.

(131) Reference is now made to FIGS. **18A** and **18B**. FIG. **18A** is a schematic illustration of another configuration of implantable structure **22**, prior to implantation, in accordance with an application of the present invention, and FIG. **18B** is a schematic illustration of implantable structure **22** in the configuration of FIG. **18A** after implantation entirely around the annulus of mitral valve **130**, before a longitudinal portion of sleeve **26** has been contracted, in accordance with an application of the present invention. In this configuration, flexible sleeve **26** is placed entirely around an annulus of mitral valve **130** in a closed loop. For some applications, sleeve **26** is introduced into left atrium **124** while first and second sleeve ends are not coupled to each other. Thereafter, in the left atrium, the sleeve is arranged entirely around the annulus to form the closed loop.

(132) Sleeve **26** is fastened to the annulus by coupling a plurality of tissue anchors **38** to the annulus. Tissue anchors **38** are coupled with: a first non-zero longitudinal density along a posterior portion of the annulus between left and right fibrous trigones **142** and **144** of the annulus, including the trigones, which density is equal to (a) a number of tissue anchors **38** coupled to the annulus

along the posterior portion of the annulus divided by (b) a length of the posterior portion of the annulus (measured along the annulus), and a second non-zero longitudinal density along an anterior portion of the annulus between left and right fibrous trigones **142** and **144** of the annulus, not including the trigones, which density is equal to (a) a number of tissue anchors **38** coupled to the annulus along the anterior portion of the annulus divided by (b) a length of the anterior portion of the annulus (measured along the annulus).

(133) The first longitudinal density is greater than the second longitudinal density. For some applications, the first longitudinal density is at least twice the second longitudinal density, such as at least 2.5 the second longitudinal density, e.g., at least 3 times the second longitudinal density. For example, tissue anchors **38A** (and, optionally **38C**) may be fastened along the posterior portion of the annulus, and tissue anchors **38B** may be fastened along the anterior portion of the annulus. After the tissue anchors are fastened to the annulus, a longitudinal portion of the sleeve is contracted, such as by causing the longitudinal contracting member to apply a force to the longitudinal portion of the sleeve, such as by actuating contracting assembly **40**.

(134) Alternatively or additionally, for some applications, sleeve **26** comprises a plurality of radiopaque markers **39**, which are positioned along the sleeve at respective longitudinal sites, such as described hereinabove with reference to FIG. **1**. The markers may provide an indication in a radiographic image (such as a fluoroscopy image) of how much of the sleeve has been deployed at any given point during an implantation procedure, in order to enable setting a desired distance between anchors **38** along the sleeve, and thus the desired differing longitudinal densities of the anchors.

(135) For some applications, as shown in FIG. **18B**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**.

(136) Reference is now made to FIG. **19**, which is a schematic illustration of implantable structure **22** after implantation around the annulus of mitral valve **130**, in accordance with an application of the present invention. In this configuration, flexible sleeve **26** is placed at least partially around an annulus of mitral valve **130**, such as partially around the annulus, as shown in FIG. **19**, or entirely around the annulus in a closed loop, such as shown in FIG. **4**, **5**, **7A-B**, **10**, **16A-B**, **17**, or **18B**, optionally using any of the techniques described herein with reference to these figures. For some applications in which the sleeve is placed entirely around the annulus, sleeve **26** is introduced into left atrium **124** while first and second sleeve ends are not coupled to each other; thereafter, in the left atrium, sleeve **26** is arranged entirely around the annulus to form the closed loop. FIG. **19** shows the annulus before a longitudinal portion of sleeve **26** has been contracted, as described below.

(137) Sleeve **26** is fastened to the annulus by coupling a plurality of tissue anchors **38** to the annulus, including first, second, and third tissue anchors **38G**, **38H**, and **38I**, as follows: one or more first tissue anchors **38G** are coupled to the annulus along a lateral scallop (**P1**) of the posterior leaflet, with a first longitudinal density, which density is equal to (a) a number of first tissue anchors **38G** coupled to the annulus along the lateral scallop (**P1**) divided by (b) a length of the lateral scallop (**P1**) along the annulus, a plurality of second tissue anchors **38H** (e.g., at least 3 tissue anchors, such as at least five tissue anchors) are coupled to the annulus along a middle scallop (**P2**) of the posterior leaflet, with a second longitudinal density, which density is equal to (a) a number of second tissue anchors **38H** coupled to the annulus along the middle scallop (**P2**) divided by (b) a length of the middle scallop (**P2**) along the annulus, and one or more third tissue anchors **38I** are coupled to the annulus along a medial scallop (**P3**) of the posterior leaflet, with a third longitudinal density, which density is equal to (a) a number of third tissue anchors **38I** coupled to the annulus along the medial scallop (**P3**) divided by (b) a length of the medial scallop (**P3**) along the annulus.

(138) Tissue anchors **38** may optionally comprise additional tissue anchors other than tissue anchors **38G**, **38H**, and **38I**, not coupled along the posterior leaflet. After the tissue anchors are fastened to the annulus, a longitudinal portion of sleeve **26** is contracted, such as by causing the longitudinal contracting member to apply a force to the longitudinal portion of the sleeve, such as by actuating contracting assembly **40**.

(139) The longitudinal densities are characterized by at least one of the following: (a) the second longitudinal density is at least twice the first longitudinal density (such as at least 2.5 the first longitudinal density, e.g., at least 3 times the first longitudinal density), and (b) the second longitudinal density is at least twice the third longitudinal density (such as at least 2.5 the third longitudinal density, e.g., at least 3 times the third longitudinal density). For some applications, both (a) the second longitudinal density is at least twice the first longitudinal density (such as at least 2.5 the first longitudinal density, e.g., at least 3 times the first longitudinal density), and (b) the second longitudinal density is at least twice the third longitudinal density (such as at least 2.5 the third longitudinal density, e.g., at least 3 times the third longitudinal density).

(140) For some applications, as shown in FIG. **19**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**.

(141) Reference is now made to FIG. **20**, which is a schematic illustration of implantable structure **22** after implantation around the annulus of mitral valve **130**, in accordance with an application of the present invention. FIG. **20** shows implantable structure **22** after a longitudinal portion of sleeve **26** has been contracted, such as by actuating contracting assembly **40**. The techniques described with reference to FIG. **20** may optionally be used in combination with the techniques described above with reference to FIG. **19**.

(142) Tissue anchors **38**, including second tissue anchors **38H**, comprise respective anchor heads **320** and tissue coupling elements **322**. Typically, anchor heads **320** are circular; alternatively, they have another shape, such as of an ellipse or a polygon (e.g., a hexagon or a square). The plurality of tissue anchors **38** are coupled to the annulus such that, after the longitudinal portion of sleeve **26** has been contracted (such as by actuating contracting assembly **40** to contract the longitudinal portion), each of anchor heads **320** of at least two of second tissue anchors **38H** coupled along the middle scallop (P2) touches at least one longitudinally-adjacent anchor head **320**; for example, each of anchor heads **320** of at least three of tissue anchors **38H** touches at least one longitudinally-adjacent anchor head **320**.

(143) Typically, before the longitudinal portion of sleeve **26** has been contracted, anchor heads **320** of the at least two of second tissue anchors **38H** do not touch any longitudinally-adjacent anchor heads **320**. Before the longitudinal portion of sleeve **26** has been contracted, the anchors are coupled to the sleeve and tissue at distances between the anchors that are less than the planned distances that the anchors move toward each other during contraction of the longitudinal portion of sleeve **26**. As a result, the anchor heads touch each other upon such contraction.

(144) By way of example, FIG. **20** shows three of tissue anchors **38H** touching at least one longitudinally-adjacent anchor head **320**. Each of the longitudinally-outer touching anchor heads touches one longitudinally-adjacent anchor head (the middle longitudinally-touching anchor head), and the middle longitudinally-touching anchor head touches two longitudinally-adjacent anchor heads (the outer touching anchor heads).

(145) This touching of longitudinally-adjacent anchor heads **320** inhibits longitudinal contraction of sleeve **26** in the longitudinal area of these anchors, so as to facilitate reshaping of the annulus in a desired manner. These longitudinally-adjacent anchor heads **320** thus are dual-function, and serve to both anchor their respective anchors to the sleeve and to inhibit contraction of the sleeve.

(146) For some applications, as shown in FIG. **20**, the plurality of tissue anchors **38** is coupled to the annulus such that, after the longitudinal portion of sleeve **26** has been contracted, such as by

causing the longitudinal contracting member to apply a force to the longitudinal portion of the sleeve, such as by actuating contracting assembly **40**: none of anchor heads **320** of first tissue anchors **38G** coupled along the lateral scallop (P1) touches any of the other anchor heads of tissue anchors **38**; and/or none of anchor heads **320** of third tissue anchors **38I** coupled along the medial scallop (P3) touches any of the other anchor heads of tissue anchors **38**.

(147) For some applications, the plurality of tissue anchors **38** are coupled to the annulus such that, after the longitudinal portion of sleeve **26** has been contracted, such as by causing the longitudinal contracting member to apply a force to the longitudinal portion of the sleeve, such as by actuating contracting assembly **40**: a first number of anchor heads **320** of first tissue anchors **38G** coupled along the lateral scallop (P1) touch at least one longitudinally-adjacent anchor head, and (b) a second number of anchor heads **320** of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, the second number greater than the first number; and/or a second number of anchor heads **320** of second tissue anchors **38H** coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, and (b) a third number of anchor heads **320** of third tissue anchors **38I** coupled along the medial scallop (P3) touch at least one longitudinally-adjacent anchor head, the second number greater than the third number.

(148) For some applications, as shown in FIG. **20**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**.

(149) Reference is now made to FIG. **21**, which is a schematic illustration of implantable structure **22** after implantation around the annulus of mitral valve **130**, in accordance with an application of the present invention. FIG. **21** shows the annulus before a longitudinal portion of sleeve **26** has been contracted, as described below. The techniques described with reference to FIG. **21** may optionally be used in combination with the techniques described hereinabove with reference to FIG. **19**, and/or the techniques described hereinabove with reference to FIG. **20**.

(150) In this configuration, flexible sleeve **26** is placed at least partially around an annulus of mitral valve **130**, such as partially around the annulus, as shown in FIG. **21**, or entirely around the annulus in a closed loop, such as shown in FIG. **4**, **5**, **7A-B**, **10**, **16A-B**, **17**, or **18B**, optionally using any of the techniques described with reference to these figures. For some applications in which the sleeve is placed entirely around the annulus, sleeve **26** is introduced into left atrium **124** while first and second sleeve ends are not coupled to each other; thereafter, in the left atrium, sleeve **26** is arranged entirely around the annulus to form the closed loop.

(151) Sleeve **26** is fastened to the annulus by coupling a plurality of tissue anchors **38** to the annulus, including tissue anchors **38J** and **38K**, such that: a first set **324** of exactly three of tissue anchors **38J** is disposed in succession along a first portion of longitudinal contracting member **30** with a first distance **D9** between longitudinal-end tissue anchors of first set **324**, measured along the annulus, and a second set **328** of exactly three of tissue anchors **38K** is disposed in succession along a second portion of longitudinal contracting member **30** with a second distance **D10** between longitudinal-end tissue anchors of second set **328**, measured along the annulus,

(152) First distance **D9** equals at least twice second distance **D10**, such as at least 2.5 times second distance **D10**, e.g., at least 3 times second distance **D10**. First distance **D9** is measured between closest portions of the longitudinal-end tissue anchors of first set **324**, and second distance **D10** is measured between closest portions of the longitudinal-end tissue anchors of second set **328**. First and second sets **324** and **328** do not share any common tissue anchors **38**. Typically, the plurality of tissue anchors **38** comprises additional tissue anchors other than tissue anchors **38J** and **38K**. After the tissue anchors are fastened to the annulus, a longitudinal portion of sleeve **26** is contracted by causing the longitudinal contracting member to apply a force to the longitudinal portion of the sleeve, such as by actuating contracting assembly **40**. Providing the greater number of anchoring

points with second set **328** better distributes forces among the anchors of this set.

(153) For some applications, as shown in FIG. **21**, implantable structure **22** is implanted with contracting mechanism **28** disposed near left fibrous trigone **142**, while for other applications (not shown, but similar to the arrangement shown in FIG. **7B**), implantable structure **22** is implanted with contracting mechanism **28** disposed near right fibrous trigone **144**.

(154) Reference is now made to FIGS. **22A-D**, which are schematic illustrations of another configuration of system **20** for repairing a dilated atrioventricular valve, and a method for deploying the system, in accordance with an application of the present invention. This configuration may be used in combination with any of the techniques and configurations described herein with reference to FIGS. **1**, **2A-I**, **3**, **19**, **20**, **21**, **24**, **25A-B**, and/or **26**.

(155) In this configuration, system **20** further comprises a linking bridge element **200**, which is configured to be coupled to sleeve **26** in order to link first and second sleeve ends **51** and **49** of sleeve **26** of implantable structure **22** via linking bridge element **200**. To this end, linking bridge element **200** typically comprises first and second bridge coupling interfaces **210A** and **210B**, which are configured to be coupled to corresponding first and second sleeve coupling interfaces **212A** and **212B** of sleeve **26**, which are disposed within 1.5 cm of first and second sleeve ends **51** and **49**, respectively, measured when the sleeve is fully longitudinally extended, such as at first and second sleeve ends **51** and **49**, respectively. For example, first and second bridge coupling interfaces **210A** and **210B** may comprise female interfaces (as shown), and first and second sleeve coupling interfaces **212A** and **212B** may comprise male interfaces (as shown), which are configured to snap into the female interfaces. Alternatively, first and second sleeve coupling interfaces **212A** and **212B** may comprise female interfaces, such as rings (e.g., comprising a metal or a plastic) integrated into the wall of sleeve **26** (configurations not shown), and first and second bridge coupling interfaces **210A** and **210B** may comprise male interfaces (configuration not shown), which are configured to snap into the female interfaces. Further alternatively, the interfaces comprise other coupling structures, as is known in the art, such as coupling structures that snap together.

(156) Typically, linking bridge element **200** has a length of at least 1 cm, no more than 5 cm, and/or between 1 and 5 cm, such as at least 1.5 cm, no more than 3.5 cm, and/or between 1.5 and 3.5 cm, e.g., 2 cm. Typically, first and second bridge coupling interfaces **210A** and **210B** are disposed within 1 cm (such as within 0.5 cm) of first and second ends **216A** and **216B** of linking bridge element **200**, respectively, e.g., between 0.5 cm and 1 cm of first and second ends **216A** and **216B** of linking bridge element **200**, respectively. For some applications, linking bridge element **200** comprises a metal or a polymer that provides longitudinal stability while maintaining some flexibility in other directions. Optionally, linking bridge element **200** further comprises a fabric or other coating for tissue growth enhancement. For some applications, linking bridge element **200** comprises elongated radial-force application element **482**, such as described hereinabove with reference to FIGS. **9-15** and/or **17**.

(157) For some applications, system **20** comprises first and second flexible longitudinal guide members **214A** and **214B**, which are removably coupled to sleeve **26** within 1.5 cm of first and second sleeve ends **51** and **49** (e.g., with 0.5 cm of the sleeve ends, or at the sleeve ends), respectively, measured when the sleeve is fully longitudinally extended. First and second flexible longitudinal guide members **214A** and **214B** extend from first and second sleeve ends **51** and **49**, respectively, away from sleeve **26**. First and second flexible longitudinal guide members **214A** and **214B** may be directly or indirectly coupled to sleeve **26**. For configurations in which first and second flexible longitudinal guide members **214A** and **214B** are indirectly coupled to sleeve **26**, the longitudinal guide members may be coupled to respective intermediary elements at locations beyond the end of the sleeve (but still within 1.5 cm of the respective sleeve ends). For example, first and second flexible longitudinal guide members **214A** and **214B** may be (a) removably coupled to first and second sleeve coupling interfaces **212A** and **212B**, respectively (in which case the longitudinal guide members may be indirectly coupled to the sleeve), and/or (b) the wall of

sleeve **26** (in which case the longitudinal guide members are directly coupled to the sleeve). For example, first and second flexible longitudinal guide members **214A** and **214B** may comprise respective sutures, wires, or strings.

(158) The longitudinal guide members are configured to guide first and second bridge coupling interfaces **210A** and **210B** to corresponding locations on sleeve **26**, such as first and second sleeve coupling interfaces **212A** and **212B**, during an implantation procedure, as shown in FIGS. **22A-C**. The longitudinal guide members removably pass through respective openings defined by linking bridge element **200**, and then through a delivery tube **220** in which linking bridge element **200** is disposed for delivery to the atrium. For some applications, the respective openings are defined by first and second bridge coupling interfaces **210A** and **210B**, respectively (as shown). For other applications, the respective openings are located elsewhere on linking bridge element **200**, typically within 10 mm, such as within 5 mm, of first and second bridge coupling interfaces **210A** and **210B**, respectively. (Optionally, longitudinal member **86**, described hereinabove with reference to FIG. **21**, also passes through delivery tube **220**.)

(159) For some applications, each of the longitudinal guide members is doubled over and threaded through its respective sleeve coupling interface and/or sleeve end. After the linking bridge element has been coupled to sleeve **26** of implantable structure **22**, the longitudinal guide members are removed by pulling on one end of each of the longitudinal guide members, typically from outside of the patient's body. Alternatively, each of the longitudinal guide members is decoupled from the sleeve in some other manner, such as using techniques described in the above-mentioned '604 application for decoupling longitudinal member **86** from contracting mechanism **40**.

(160) Typically, as described hereinabove, implantable structure **22** comprises longitudinal contracting member **30**, which is configured to longitudinally contract a longitudinal portion of sleeve **26**, as described hereinabove. Longitudinal contracting member **30** may be disposed with respect to the sleeve in any of the arrangements described hereinabove, including those regarding the extent to which the contracting member extends along the length of sleeve. First and second flexible longitudinal guide members **214A** and **214B** are separate and distinct from longitudinal contracting member **30**; in other words, first and second flexible longitudinal guide members **214A** and **214B** are not fixed to longitudinal contracting member **30**, and are not parts of a common longitudinal member.

(161) Typically, when first and second flexible longitudinal guide members **214A** and **214B** are removably coupled to sleeve **26** of implantable structure **22**: no portion of either first flexible longitudinal guide member **214A** or second flexible longitudinal guide member **214B** is disposed more than 1.5 cm from first and second sleeve ends **51** and **49**, respectively, measured when the sleeve is fully longitudinally extended; first and second flexible longitudinal guide members **214A** and **214B** are collectively disposed along less than 30% of a length of sleeve **26**, such as less than 5% of the length of the sleeve, measured when the sleeve is fully longitudinally extended; and/or for applications in which implantable structure **22** comprises longitudinal contracting member **30**, first and second flexible longitudinal guide members **214A** and **214B** do not longitudinally overlap longitudinal contracting member **30** (i.e., are not disposed at any common longitudinal locations with longitudinal contracting member **30**).

(162) Alternatively, for some applications, system **20** comprises a single flexible longitudinal guide member **214** which removably passes through the entire sleeve **26** (configuration not shown). After the linking bridge element has been coupled to sleeve **26** of implantable structure **22**, the longitudinal guide member is removed by pulling on one end of the longitudinal guide member, typically from outside of the patient's body. Alternatively, the longitudinal guide member is decoupled from the sleeve in some other manner, such as using techniques described in the above-mentioned '604 application for decoupling longitudinal member **86** from contracting mechanism **40**.

(163) After first and second bridge coupling interfaces **210A** and **210B** have been guided over first

and second flexible longitudinal guide members **214A** and **214B** to corresponding first and second sleeve coupling interfaces **212A** and **212B**, as shown in FIGS. **22A-C**, first and second bridge coupling interfaces **210A** and **210B** are coupled to corresponding first and second sleeve coupling interfaces **212A** and **212B**, also as shown in FIG. **24C**. For example, first and second tubes **222A** and **222B** may be introduced through delivery tube **220** and over first and second flexible longitudinal guide members **214A** and **214B**, respectively, and used to push the corresponding coupling interfaces against each other, until they snap together, as shown in FIG. **22C**.

(164) FIG. **22D** shows linking bridge element **200** coupled to sleeve **26** of implantable structure **22**, after the delivery tool has been removed from the atrium.

(165) For applications in which implantable structure **22** comprises longitudinal contracting member **30**, the implantation method typically comprises: during a percutaneous transcatheter procedure, placing sleeve **26** of implantable structure **22** partially around an annulus of a valve of a subject, such as a mitral valve or tricuspid valve (typically around all or a portion of a posterior portion of the annulus between fibrous trigones of the valve); anchoring sleeve **26** to cardiac tissue, such as described hereinabove with reference to FIGS. **2G-I**; coupling linking bridge element **200** to sleeve **26**, as described hereinabove, typically along all or a portion of an anterior portion of the annulus between the fibrous trigones; and thereafter, contracting a longitudinal portion of sleeve **26** by causing longitudinal contracting member **30** to apply a contracting force to the longitudinal portion of the sleeve, as described hereinabove.

(166) Thus, the contracting of the sleeve is not performed simultaneously with the coupling of the linking bridge element to the sleeve. Moreover, longitudinal contracting member **30** does not serve as either of first and second flexible longitudinal guide members **214A** and **214B**.

(167) Optionally, for some applications, system **20** comprises one or more bridge anchors **224** (e.g., one, two, or three bridge anchors **224**), which are used to couple linking bridge element **200** to tissue at the anterior portion of the annulus. For some applications, the one or more bridge anchors **224** are deployed using anchor deployment manipulator **24**, described hereinabove.

(168) Reference is now made to FIGS. **23A-B**, which are schematic illustrations of another configuration of linking bridge element **200**, in accordance with an application of the present invention. Other than as described below, this configuration is identical to the configuration described hereinabove with reference to FIGS. **22A-D**.

(169) In this configuration, first and second bridge coupling interfaces **210A** and **210B** are male interfaces, which are configured to pierce the wall of sleeve **26**, thereby becoming coupled to the sleeve. For example, the coupling elements may be shaped as harpoons or other barbed structures. In this configuration, sleeve **26** typically does not comprise any coupling interfaces or coupling elements.

(170) Reference is now made to FIG. **24**, which is a schematic illustration of contracting mechanism **28**, disassembled to show a relationship among individual components of the contracting mechanism, in accordance with an application of the present invention. The components are arranged and function as described with reference to FIG. **7** of the above-mentioned '604 publication, *mutatis mutandis*.

(171) Reference is made to FIGS. **25A-B** and **26**, which are schematic illustrations of a valve prosthesis assembly **400**, in accordance with respective applications of the present invention. Valve prosthesis assembly **400** comprises a prosthetic heart valve **410** that is couplable to a base ring **422**. Prosthetic heart valve **410** is used to replace a native diseased heart valve. Valve **410** comprises a plurality of artificial leaflets **430**, which comprise a pliant material. Valve **410** may implement techniques known in the artificial valve art, such as described, for example, in US Patent Application Publication 2007/0255400 to Parravicini et al., US Patent Application Publication 2004/0122514 to Fogarty et al., US Patent Application Publication 2007/0162111 to Fukamachi et al., and/or US Patent Application Publication 2008/0004697 to Lichtenstein et al., all of which are incorporated herein by reference.

(172) Valve **410** further comprises an annular base **432**, to which artificial leaflets **430** are coupled. Annular base **432** is configured to be couplable to base ring **422** during an implantation procedure. For example, as shown in FIG. **26**, base ring **422** may comprise one or more coupling elements **434**, such as clips or magnets, which are configured to be coupled to corresponding coupling elements on a lower surface of annular base **432** (not visible in the figures). Alternatively or additionally, annular base **432** may be configured to be placed within the opening defined by base ring **422**, as shown in FIG. **25A**. To hold the annular base coupled to the base ring, the base ring is tightened around the annular base, as shown in FIG. **25B**, typically using one or more of the techniques described hereinabove for contracting implantable structures. Typically, valve prosthesis assembly **400**, such as annular base **432** thereof, is configured to push and hold open the intact diseased native leaflets.

(173) Base ring **422** implements one or more of the techniques of implantable structure **22** described hereinabove. In particular, base ring **422** may be coupled to the annulus of the native diseased valve using the anchoring techniques described hereinabove. In addition, base ring **422** typically comprises sleeve **26** and contracting mechanism **28**, which may, for some applications, comprise a rotatable structure, such as a spool **46**, which is typically implemented using techniques described herein. The contracting mechanism is arranged to contract base ring **422**, e.g., the rotatable structure is arranged such that rotation thereof contracts base ring **422**, typically using techniques described herein. Such tightening may serve to couple base ring **422** to annular base **432**, as shown in FIG. **25B**. Alternatively or additionally, such tightening sets the desired dimensions of the base ring, in order to align the coupling elements of the base ring with those of valve **410**, thereby enabling tight coupling, such as for the applications described with reference to FIG. **26**.

(174) For some applications, as shown in FIG. **26**, base ring **422** comprises a partial ring, such as described hereinabove with reference to FIGS. **2A-I**, **19**, **20**, and **21**. For other applications, as shown in FIGS. **25A-B**, the base ring is arranged as a full ring, such as described hereinabove with reference to FIGS. **4**, **5**, **7A-B**, **10**, **16A-B**, **17**, and **18B**.

(175) Valve prosthesis assembly **400** is typically implanted in a minimally invasive transcatheter or percutaneous procedure. The procedure begins with the introduction and implantation of base ring **422** into the heart, such as using techniques for implanting implantable structure **22**, described hereinabove with reference to FIGS. **2A-I**. Prosthetic heart valve **410** is subsequently introduced into the heart and coupled to base ring **422**, as described above. Valve prosthesis assembly **400** is typically used for replacement of a diseased native mitral valve, aortic valve, tricuspid valve, or pulmonary valve.

(176) For some applications, system **20** further comprises a closure mechanism, such as described in above-mentioned US Patent Application Publication 2012/0330411, with reference to FIGS. **16-17B** thereof.

(177) For some applications, system **20** further comprises a flexible pusher element, such as described and shown in US Patent Application Publication 2010/0286767, which is incorporated herein by reference, with reference to FIG. **8** thereof. The pusher element aids with accurately positioning successive anchors **38** during an implantation procedure, such as described hereinabove with reference to FIGS. **2H** and **21**. For some applications, system **20** further comprises a pusher tube that is applied to proximal end **49** of sleeve **26**, such as described in the above-mentioned '604 publication, with reference to FIGS. **14** and/or **18A-B** thereof. For some applications, system **20** further comprises a steerable tube, such as described in the above-mentioned '604 publication, with referenced to FIG. **15** thereof, or with reference to FIG. **16** thereof. For some applications, system **20** further comprises a pulling wire, such as described in the above-mentioned '604 publication, with referenced to FIG. **17** thereof. For some applications, system **20** further comprises an external control handle, such as described in the above-mentioned '604 publication, with referenced to FIG. **19** thereof. For some applications, contracting assembly **40** and implantable structure **22** are

configured as described with reference to FIG. 23 of the above-mentioned '604 publication, mutatis mutandis.

(178) For some applications of the present invention, system **20** is used to treat an atrioventricular valve other than the mitral valve, i.e., the tricuspid valve. For these applications, implantable structure **22** and other components of system **20** described hereinabove as being placed in the left atrium are instead placed in the right atrium.

(179) Although implantable structure **22** is described hereinabove as being placed in an atrium, for some application the implantable structure is instead placed in either the left or right ventricle.

(180) The scope of the present invention includes applications described in the following applications, which are incorporated herein by reference. In an application, techniques and apparatus described in one or more of the following applications are combined with techniques and apparatus described herein: PCT Publication WO 06/097931 to Gross et al., entitled, "Mitral Valve treatment techniques," filed Mar. 15, 2006; U.S. Provisional Patent Application 60/873,075 to Gross et al., entitled, "Mitral valve closure techniques," filed Dec. 5, 2006; U.S. Provisional Patent Application 60/902,146 to Gross et al., entitled, "Mitral valve closure techniques," filed on Feb. 16, 2007; U.S. Provisional Patent Application 61/001,013 to Gross et al., entitled, "Segmented ring placement," filed Oct. 29, 2007; PCT Patent Application PCT/IL07/001503 to Gross et al., entitled, "Segmented ring placement," filed on Dec. 5, 2007, which published as PCT Publication WO 08/068756; U.S. patent application Ser. No. 11/950,930 to Gross et al., entitled, "Segmented ring placement," filed on Dec. 5, 2007, which published as US Patent Application Publication 2008/0262609; U.S. Provisional Patent Application 61/132,295 to Gross et al., entitled, "Annuloplasty devices and methods of delivery therefor," filed on Jun. 16, 2008; U.S. patent application Ser. No. 12/341,960 to Cabiri, entitled, "Adjustable partial annuloplasty ring and mechanism therefor," filed on Dec. 22, 2008, which published as US Patent Application Publication 2010/0161047; U.S. Provisional Patent Application 61/207,908 to Miller et al., entitled, "Actively-engageable movement-restriction mechanism for use with an annuloplasty structure," filed on Feb. 17, 2009; U.S. patent application Ser. No. 12/435,291 to Maisano et al., entitled, "Adjustable repair chords and spool mechanism therefor," filed on May 4, 2009, which published as US Patent Application Publication 2010/0161041; U.S. patent application Ser. No. 12/437,103 to Zipory et al., entitled, "Annuloplasty ring with intra-ring anchoring," filed on May 7, 2009, which published as US Patent Application Publication 2010/0286767; PCT Patent Application PCT/IL2009/000593 to Gross et al., entitled, "Annuloplasty devices and methods of delivery therefor," filed on Jun. 15, 2009, which published as PCT Publication WO 10/004546; U.S. patent application Ser. No. 12/548,991 to Maisano et al., entitled, "Implantation of repair chords in the heart," filed on Aug. 27, 2009, which published as US Patent Application Publication 2010/0161042; U.S. patent application Ser. No. 12/608,316 to Miller et al., entitled, "Tissue anchor for annuloplasty ring," filed on Oct. 29, 2009, which published as US Patent Application Publication 2011/0106247; U.S. Provisional Patent Application 61/265,936 to Miller et al., entitled, "Delivery tool for implantation of spool assembly coupled to a helical anchor," filed Dec. 2, 2009; PCT Patent Application PCT/IL2009/001209 to Cabiri et al., entitled, "Adjustable annuloplasty devices and mechanisms therefor," filed on Dec. 22, 2009, which published as PCT Publication WO 10/073246; U.S. patent application Ser. No. 12/689,635 to Zipory et al., entitled, "Over-wire rotation tool," filed on Jan. 19, 2010, which published as US Patent Application Publication 2010/0280604; U.S. patent Ser. No. 12/689,693 to Hammer et al., entitled, "Deployment techniques for annuloplasty ring," filed on Jan. 19, 2010, which published as US Patent Application Publication 2010/0280605; U.S. patent application Ser. No. 12/706,868 to Miller et al., entitled, "Actively-engageable movement-restriction mechanism for use with an annuloplasty structure," filed on Feb. 17, 2010, which published as US Patent Application Publication 2010/0211166; PCT Patent Application PCT/IL2010/000357 to Maisano et al., entitled, "Implantation of repair chords in the heart," filed May 4, 2010, which published as PCT Publication WO 10/128502; PCT Patent

Application PCT/IL2010/000358 to Zipory et al., entitled, "Deployment techniques for annuloplasty ring and over-wire rotation tool," filed May 4, 2010, which published as PCT Publication WO 10/128503; and/or U.S. patent application Ser. No. 13/167,476 to Hammer et al., filed Jun. 23, 2011, entitled, "Closure element for use with an annuloplasty structure," which published as US Patent Application Publication 2012/0330410.

(181) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. A method for use with a plurality of tissue anchors, each of the tissue anchors having a respective anchor head, the method comprising: (1) during a percutaneous transcatheter procedure, securing an annuloplasty structure at least partially around an annulus of a mitral valve of a subject by coupling the plurality of tissue anchors of the annuloplasty structure to the annulus, such that (i) a longitudinal contracting member of the annuloplasty structure extends at least partially around the annulus, and (ii) the plurality of tissue anchors is distributed at least partially around the annulus, with: a first longitudinal density of the tissue anchors along a lateral scallop (P1) of a posterior leaflet of the mitral valve, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the lateral scallop (P1) divided by (b) a length of the lateral scallop (P1) along the annulus, a second longitudinal density of the tissue anchors along a middle scallop (P2) of the posterior leaflet, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the middle scallop (P2) divided by (b) a length of the middle scallop (P2) along the annulus, and a third longitudinal density of the tissue anchors along a medial scallop (P3) of the posterior leaflet, which density is equal to (a) a number of the tissue anchors coupled to the annulus along the medial scallop (P3) divided by (b) a length of the medial scallop (P3) along the annulus, wherein the second longitudinal density is at least twice as great as at least one of the following: (a) the first longitudinal density, and (b) the third longitudinal density; and (2) thereafter, contracting at least part of the annulus by contracting a longitudinal portion of the annuloplasty structure by tightening the longitudinal contracting member, the second longitudinal density being sufficiently great that the tightening causes contact between adjacent anchor heads of at least two of the anchors coupled to the annulus along the middle scallop.

2. The method according to claim 1, wherein the annuloplasty structure includes a locking mechanism, and wherein the method further comprises, after step (2), locking the longitudinal contracting member with respect to the locking mechanism.

3. The method according to claim 2, wherein the annuloplasty structure includes a contracting assembly, and wherein the tightening, in step (2), of the longitudinal contracting member comprises tightening the longitudinal contracting member by actuating the contracting assembly.

4. The method according to claim 1, wherein both (a) the second longitudinal density is at least twice the first longitudinal density, and (b) the second longitudinal density is at least twice the third longitudinal density.

5. The method according to claim 1, wherein the coupling, in step (1), of the plurality of tissue anchors to the annulus comprises coupling the plurality of tissue anchors to the annulus such that, before step (2), the adjacent anchor heads of the at least two of the anchors do not touch.

6. The method according to claim 1, wherein the coupling, in step (1), of the plurality of tissue anchors to the annulus comprises coupling the plurality of tissue anchors to the annulus such that, after step (2), (a) none of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touches any of the other anchor heads of the tissue anchors, and (b) none of the anchor heads

of the tissue anchors coupled along the medial scallop (P3) touches any of the other anchor heads of the tissue anchors.

7. The method according to claim 1, wherein the coupling, in step (1), of the plurality of tissue anchors to the annulus comprises coupling the plurality of tissue anchors to the annulus such that, after step (2), (a) a first number of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touch at least one longitudinally-adjacent anchor head, and (b) a second number of the anchors heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, the second number greater than the first number.

8. The method according to claim 1, wherein the coupling, in step (1), of the plurality of tissue anchors to the annulus comprises coupling the plurality of tissue anchors to the annulus such that, after step (2), (a) a second number of the anchor heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, and (b) a third number of the anchors heads of the tissue anchors coupled along the medial scallop (P3) touch at least one longitudinally-adjacent anchor head, the second number greater than the third number.

9. The method according to claim 1, wherein the coupling, in step (1), of the plurality of tissue anchors to the annulus comprises coupling the plurality of tissue anchors to the annulus such that, after step (2): a first number of the anchor heads of the tissue anchors coupled along the lateral scallop (P1) touch at least one longitudinally-adjacent anchor head, a second number of the anchors heads of the tissue anchors coupled along the middle scallop (P2) touch at least one longitudinally-adjacent anchor head, and a third number of the anchors heads of the tissue anchors coupled along the medial scallop (P3) touch at least one longitudinally-adjacent anchor head, the second number greater than the first number, and the second number greater than the third number.

10. The method according to claim 1, wherein the annuloplasty structure has first and second ends, and wherein the method comprises introducing the annuloplasty structure into a left atrium of the subject while the first and the second ends are not coupled to each other.

11. The method according to claim 10, wherein step (1) comprises securing the annuloplasty structure entirely around the annulus to form a closed loop, after the introducing of the annuloplasty structure into the left atrium while the first and the second ends are not coupled to each other.

12. The method according to claim 1, wherein the annuloplasty structure is shaped so as to define an integrally closed loop having no ends, and wherein the method comprises introducing, into a left atrium of the subject, the annuloplasty structure that is shaped so as to define the integrally closed loop having no ends.

13. A method comprising: during a percutaneous transcatheter procedure, securing an annuloplasty structure at least partially around an annulus of a mitral valve of a subject by coupling a plurality of tissue anchors of the annuloplasty structure to the annulus, such that (i) a longitudinal contracting member of the annuloplasty structure extends at least partially around the annulus, and (ii) the plurality of tissue anchors is distributed at least partially around the annulus, with: a first subset of the tissue anchors distributed along a lateral scallop (P1) of a posterior leaflet of the mitral valve, a second subset of the tissue anchors distributed along a middle scallop (P2) of the posterior leaflet, and a third subset of the tissue anchors distributed along a medial scallop (P3) of the posterior leaflet, wherein the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of an other subset selected from the group consisting of the first subset, and the third subset; and thereafter, contracting the annulus by drawing anchors of the plurality toward each other by tightening the longitudinal contracting member such that (i) proximity between anchors of the second subset obstructs the anchors of the second subset from being drawn toward each other, whereas (ii) proximity between anchors of the other subset facilitates the anchors of the other subset being drawn toward each other.

14. A method comprising: during a percutaneous transcatheter procedure, securing an annuloplasty structure at least partially around an annulus of a mitral valve of a subject by coupling a plurality of

tissue anchors of the annuloplasty structure to the annulus, such that (i) a longitudinal contracting member of the annuloplasty structure extends at least partially around the annulus, and (ii) the plurality of tissue anchors is distributed at least partially around the annulus, with: a first subset of the tissue anchors distributed along a lateral scallop (P1) of a posterior leaflet of the mitral valve, a second subset of the tissue anchors distributed along a middle scallop (P2) of the posterior leaflet, and a third subset of the tissue anchors distributed along a medial scallop (P3) of the posterior leaflet, wherein the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of an other subset selected from the group consisting of the first subset, and the third subset; and thereafter, contracting the annulus by drawing anchors of the plurality toward each other by tightening the longitudinal contracting member such that contact between anchors of the second subset obstructs the anchors of the second subset from being drawn toward each other.

15. The method according to claim 13, wherein the annuloplasty structure includes a locking mechanism, and wherein the method further comprises, after the contracting of the annulus, locking the longitudinal contracting member with respect to the locking mechanism.

16. The method according to claim 13, wherein both (a) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the first subset, and (b) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the third subset.

17. The method according to claim 13, wherein at least one of either (a) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the first subset, and (b) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the third subset.

18. The method according to claim 13, wherein the coupling of the plurality of tissue anchors to the annulus comprises coupling at least 3 tissue anchors to the annulus along the middle scallop.

19. The method according to claim 13, wherein the annuloplasty structure has first and second ends, and wherein the method comprises introducing the annuloplasty structure into a left atrium of the subject while the first and the second ends are not coupled to each other.

20. The method according to claim 19, wherein the securing comprises securing the annuloplasty structure entirely around the annulus to form a closed loop, after the introducing of the annuloplasty structure into the left atrium while the first and the second ends are not coupled to each other.

21. The method according to claim 13, wherein the annuloplasty structure is shaped so as to define an integrally closed loop having no ends, and wherein the method comprises introducing, into a left atrium of the subject, the annuloplasty structure that is shaped so as to define the integrally closed loop having no ends.

22. The method according to claim 14, wherein the annuloplasty structure includes a locking mechanism, and wherein the method further comprises, after the contracting of the annulus, locking the longitudinal contracting member with respect to the locking mechanism.

23. The method according to claim 22, wherein the annuloplasty structure includes a contracting assembly, and wherein the tightening of the longitudinal contracting member comprises tightening the longitudinal contracting member by actuating the contracting assembly.

24. The method according to claim 14, wherein both (1) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the first subset, and (2) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the third subset.

25. The method according to claim 14, wherein at least one of (1) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the first subset, and (2) the anchors of the second subset are distributed at least twice as close to each other compared with the anchors of the third subset.

26. The method according to claim 14, wherein the coupling of the plurality of tissue anchors to the annulus comprises coupling at least 3 tissue anchors to the annulus along the middle scallop.
27. The method according to claim 14, wherein the annuloplasty structure has first and second ends, and wherein the method comprises introducing the annuloplasty structure into a left atrium of the subject while the first and the second ends are not coupled to each other.
28. The method according to claim 27, wherein the securing comprises securing the annuloplasty structure entirely around the annulus to form a closed loop, after the introducing of the annuloplasty structure into the left atrium while the first and the second ends are not coupled to each other.
29. The method according to claim 14, wherein the annuloplasty structure is shaped so as to define an integrally closed loop having no ends, and wherein the method comprises introducing, into a left atrium of the subject, the annuloplasty structure that is shaped so as to define the integrally closed loop having no ends.
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